

«description»

Rev	ECN No.	Date	Reviewed By	Approved By	Status
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Summary:

This procedure specifies the method for installing the ROV-Operated Mechanical Tree Running Tool ([P1000170237](#)).

It is not advised to paste this procedure directly into any Project Specific XT Installation Procedure. Any enhancements made to this specific procedure would be difficult to implement across multiple project procedures.

The tool can be re-configured from a Tree Handling/Test Tool (XTHT) to a Mechanical Tree Running Tool (MTRT) by removing the two external s-seals (DBI items 20 & 21 on Assembly DWG [DU100239949](#)).

This procedure is only approved for use when using the MTRT with a completed tree. Any of retrievable components of the XT it's not installed, separate calculations of CoG has to be done prior to operation.

REV B OF THIS PROCEDURE HAS BEEN REVIEWED AND RELEASED FOR TESTING BY HOUSTON, USING CHECKLIST [LST10136228](#) REV B.

REV C OF THIS DOCUMENT DO NOT INCLUDE ANY CHANGES TO PRESSURE TESTING PROCEDURES THUS LST10136228 IS STILL VALID.

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Abbreviations

The following abbreviations are used throughout this procedure.

Abbreviation	Definition
"L"	Lock
"U"	Unlock
ADF	Assembly Documentation File
API	American Petroleum Institute
ASSY	Assembly
CCW	Counter-clockwise (rotation)
COG	Center of Gravity
CPY	Company
CVB	Composite Valve Block
CW	Clockwise (rotation)
DBI	Database Information (for a Part)
DWG	Drawing
ECN	Engineering Change Notice
EVDT	Enhanced Vertical Deepwater Tree
HCLS	Heave Compensated Landing System
ID	Inner Diameter
JSA	Job Safety Analysis
k, ksi, kips	Kilo-Pounds per Square In = 1000 lbf
MTRT	Mechanical Tree Running Tool
NA	Not Applicable
OD	Outer Diameter
REG	Regular
REV	Revision
ROV	Remote Operated Vehicle
XT	Christmas Tree
XTHT	Christmas Tree Handling Tool/Test Tool

Safety Notice



Prior to performing any work requiring pressure, you shall read and understand the applicable safety notifications within [Global Practice Standard](#). You shall then sign the Safety Notice Data Collection Table.

WARNING

1.0 Safety-Notice - Data Collection Table

Append your name to the list below, after you have read and understood the applicable safety notifications in the document and Global Practice Standard .			
Name (Printed)	Date	Global Practice Standard Revision	Initial

2.0 Symbols and Conventions

The following words and symbols found throughout this manual mark special messages to alert the operator of specific information concerning the PERSONNEL, the EQUIPMENT or the PROCESS.



Text set off in this manner provides warning notice that failure to follow these directions in this **WARNING** can result in bodily harm or loss of life and/or extensive damage to equipment.

WARNING



Text set off in this manner provides warning notice that failure to follow these directions in this **CAUTION** can result in damage to equipment.

CAUTION



Text set off in this manner present clarifying information or specific instructions pertinent to the immediate instruction.

NOTE

3.0 JSA Sign-off for MTRT Installation

A JSA must be completed and approved by authorized service personnel only. All personnel performing this operation must completely comprehend, understand, and sign the JSA. At no time shall anyone work on the operation without reading and signing the JSA.



DO NOT PROCEED UNTIL THIS STEP HAS BEEN COMPLETED.

WARNING



Signoffs have been strategically placed throughout the procedure to highlight critical and logical steps. The absence of a signature block does not permit skipping of a step in the procedure.

NOTE

4.0 MTRT Information

4.1 Reference Documentation



The MTRT Assembly listed below is the assembly that will actually be installed into the XT and used for Running/Retrieving the XT. Refer to [DU100239949](#).

NOTE

MTRT Assembly (with Shipping Stand)

- [P1000171873](#) - DBI
- [DU100245944](#) - Drawing

MTRT Assembly (without Shipping Stand)

- [P1000170237](#) - DBI
- [DU100239949](#) - Drawing

MTRT Shipping Stand

- [P1000170649](#) - DBI
- [DU100245966](#) - Drawing

Reference Documents and Drawings

- [DU100129866](#) - Layout, MTRT/XTHT in XT
- [DU100128635](#) - Layout, MTRT/XTHT Lock/Unlock, Installation Sequence
- [DU100162251](#) - Layout, MTRT/XTHT Counterweight Configurations
- [ASY10131969](#) - Assembly Procedure, 15k MTRT/XTHT
- [THI10131995](#) - Transport & Handling Instructions, 15k MTRT/XTHT
- [TST10057864](#) - Test Procedure , 15k MTRT/XTHT
- [PSM10060305](#) - Preservation, Storage And Maintenance Procedure, 15k MTRT/XTHT

4.2 Reference Cross-Section

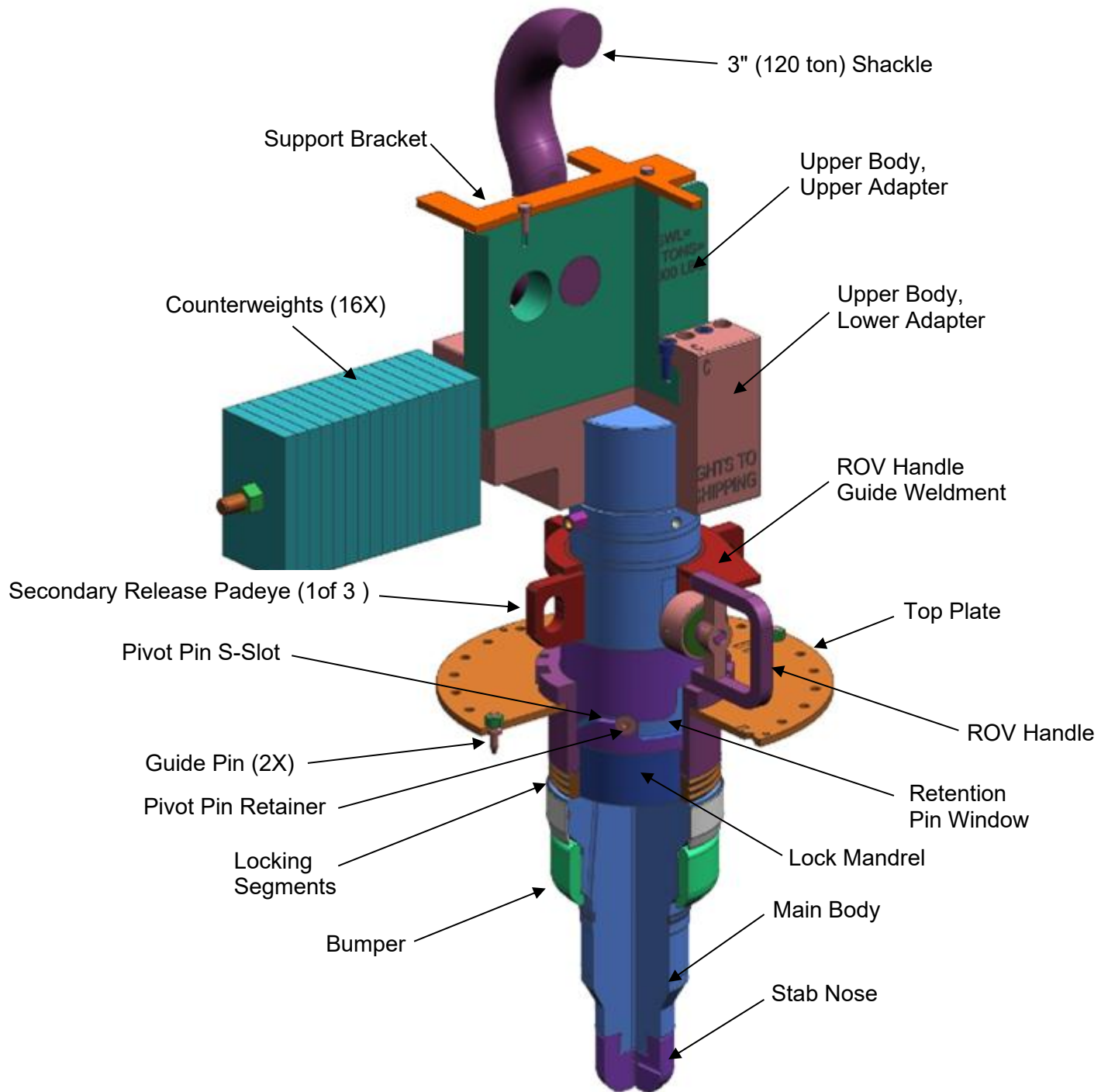


Figure 1: 3-D Cross-Section of MTRT / XTHT Showing Internal Components

4.3 Running Methods

There are two methods for running the MTRT: Wireline or Drillpipe.



NOTE

The Customer is responsible for ensuring that all certifications for any non-FMC equipment above the MTRT, when used in conjunction with the tool, meet the criteria required for offshore installation equipment.

4.3.1 Wireline

- This wireline can be from a winch on the rig or from a rig-winch, heave comp crane, or HCLS-type system. The rigging is attached directly to the 3" 120 ton shackle on the MTRT.
- Use of swivel (supplied by the customer) is required to be used in the rigging immediately above the tool.



WARNING

If swivel is not used in the rigging directly above the MTRT, it is extremely important that selected rigging configuration be oriented directly above the lift point on the MTRT and within alignment to less than 5 degrees. Ensure to release any torsion build up in the rigging configuration prior to performing any lift. Ensure torque build-up due to rigging or external effects does not exceed 12,500 ft-lbs.

Failure to follow these guidelines can result in MTRT being subjected to excessive torque.

4.3.2 Drillpipe

- Drillpipe must be adapted to 6-5/8 API REG pin.
- Make sure that box/box 6-5/8 API REG pin its available on the rig (CPY supply) for making up to TFMC adapter [P1000003329](#).
- Use TFMC 6-5/8 API REG pin X padeye ([P1000003329](#)) for loads up-to 85 metric tonnes with the 3" 120 ton shackle ([P122864-0000](#)). To connect this shackle to the 3" 120 ton shackle on the MTRT, use a single point sling (project specific).



Ensure that the sling being used is fit for the project specific XT with respect to the XT weight for both offshore and onshore use.

NOTE

- Use of swivel (supplied by the customer) is required to be used in the rigging immediately above the tool.



If swivel is not used in the rigging directly above the MTRT, it is extremely important that selected rigging configuration be oriented directly above the lift point on the MTRT and within alignment to less than 5 degrees. Ensure to release any torsion build up in the rigging configuration prior to performing any lift. Ensure torque build-up due to rigging or external effects does not exceed 12,500 ft-lbs.

WARNING

Failure to follow these guidelines can result in MTRT being subjected to excessive torque.



MTRT shall be lifted using the 3" 120 Ton shackle at all times. Alternatively, Swivel Hoist Rings (2x) may be used for lifting and handling MTRT assembly (without XT) only for MTRT offset settings of 3.0" or less. MTRT offset settings of 3.0" or less do not require the use of any counterweights.

NOTE

When handling the MTRT assembly with no counterweights or half the number of counterweights, park the remaining counterweights on the shipping stand. Refer to [DU100162251](#) for relationship between number of counterweights to be used and corresponding MTRT offset settings.

5.0 MTRT Configuration



Section 5.0 is to be performed on the MTRT regardless of which operation (Running XT, Retrieving XT, or Handling XT) is being performed.

NOTE



If using the MTRT with 10K EVDT XTs, remove the guide pins and the respective nuts from the tool assembly. The 10K EVDT XTs do not have drilled and tapped holes on the top that would interface with the Guide Pins and so leaving them in place would prevent the MTRT from locking.

CAUTION

The MTRT must be installed into the Shipping Stand (P1000170649) anytime it is not Locked to the XT.

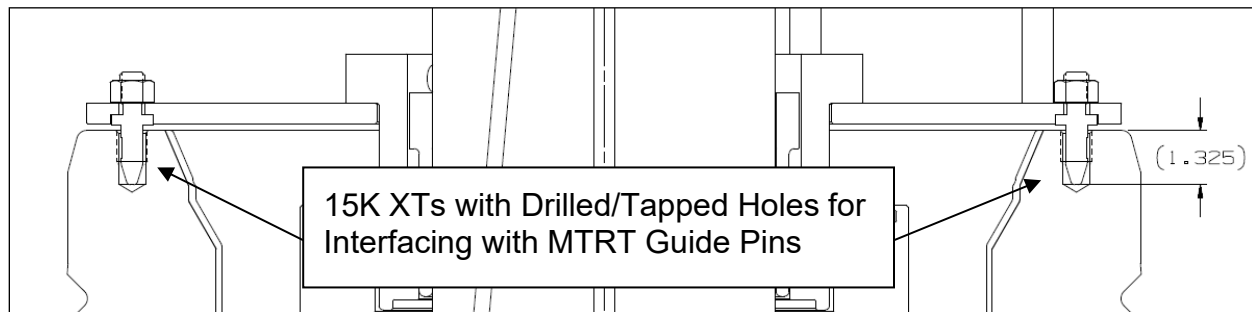


Figure 2: 15K XT with Drilled/Tapped Holes on the Top (Holes Absent on 10K XTs)

5.1 Determine Required Offsets

1. Consult the drawing of the XT that is to be handled and locate the COG from the top view of the tree, as shown in Figure 3.
 - The "front" of the tree is considered to be the side with the master valve actuator.
 - The MTRT has two offset adjustments, a linear offset and an angular offset.

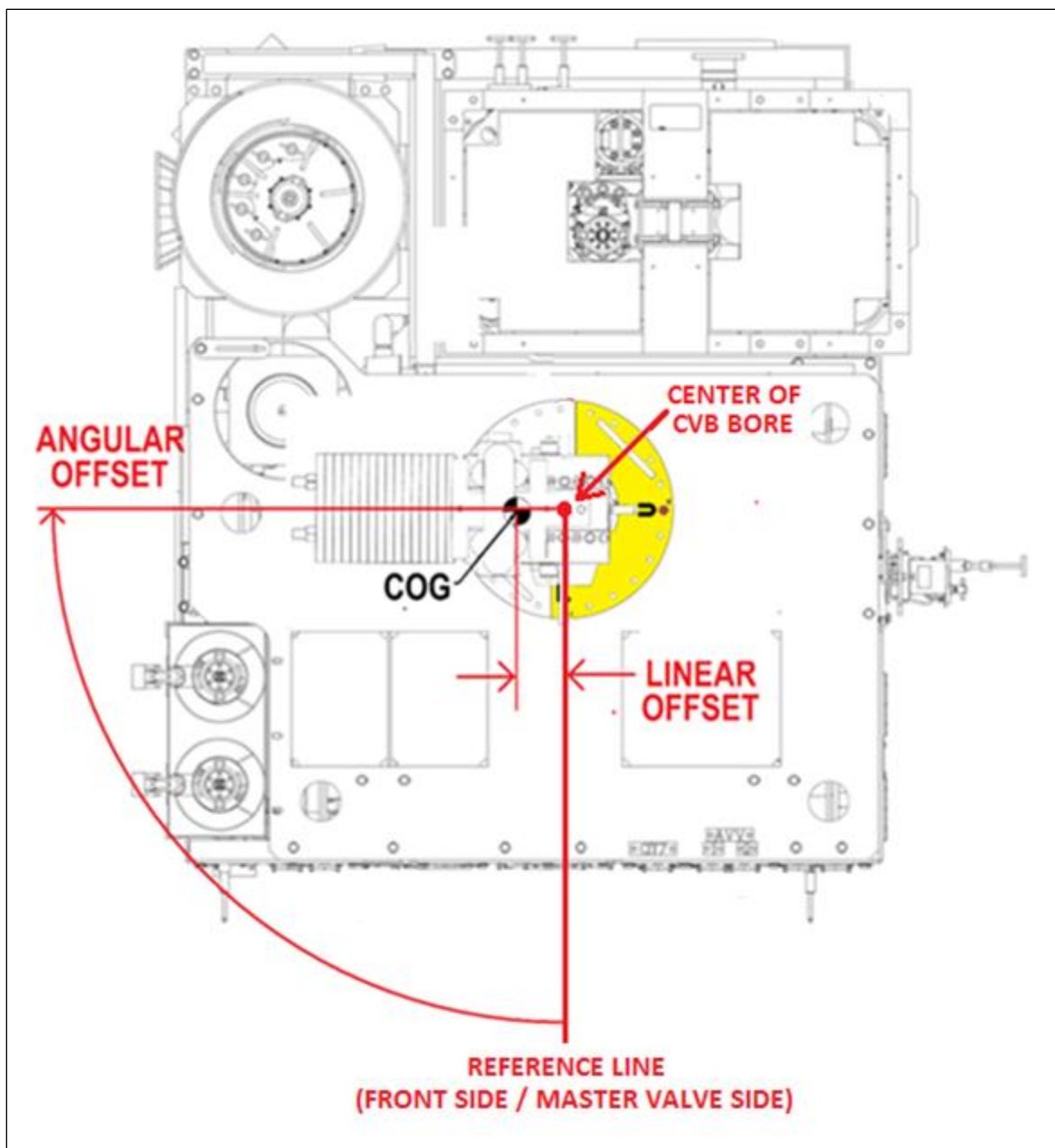


Figure 3: Top View of MTRT Installed in EVDT Tree (Reference Only)

2. Record the required LINEAR offset.

- See Figure 3.
- This is the measurement from the CVB centerline to the XT COG.

LINEAR OFFSET = _____ Inches

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3. Record the required ANGULAR offset.

- See Figure 3.
- This is the measurement from the Reference Line to the COG going CLOCKWISE.
- The reference line is always oriented towards the FRONT of the XT, which is the Main ROV Panel / Master Valve side.
- There will typically be a black vertical line painted on the CVB, but this does not necessarily indicate the COG orientation. The drawing of the XT should be checked to determine the COG orientation.

ANGULAR OFFSET = _____ Degrees

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NOTE

The Linear Offset value will be used in Section 0

Configure Linear Offset and the Angular Offset value will be used in Section 5.3.

5.2 Configure Linear Offset



WARNING

While installing the MTRT shackle, careful consideration should be made to the overall weight of the shackle and pin assembly. Appropriate number of personnel should be tasked to perform the step above to prevent dropped objects and pinch points while installing the shackle into the MTRT Upper Body Upper Adapter. Shackle weight is 178 lbs.

5.2.1 Shackle Positioning

1. If required, move the 120 Ton Shackle to the correct hole/position (reference Figure 4). Note that the Upper Adapter position may need to be adjusted and the shackle may have to be left off until this is done so that fasteners are not blocked:
 - Use the "A" Position if the linear offset needs to be adjusted between 5 and 7.5 inches.
 - Use the "B" Position if the linear offset needs to be adjusted between 0 and 4.5 inches.

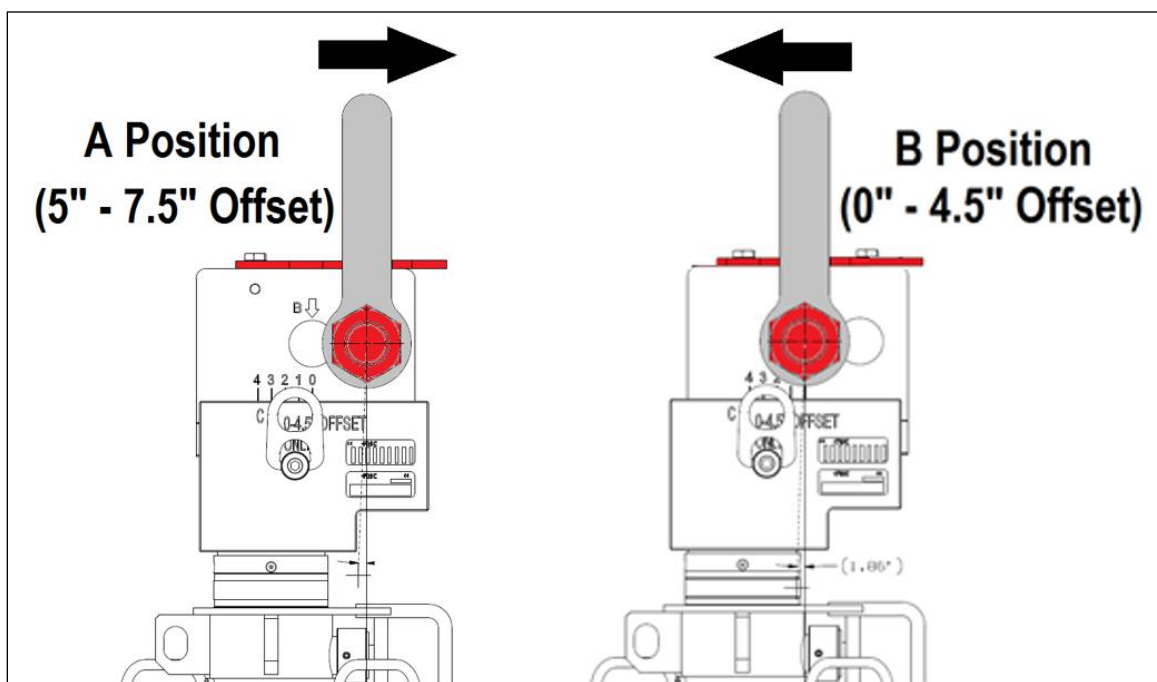


Figure 4: Shackle Position

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2. To adjust the linear offset, remove Qty. 6 fasteners from either side of the Lower Adapter (item 2) as shown below. Note that the shackle may need to be left off until the fasteners are reinstalled if shackle will be in the "B" position. See right side of Figure 5.

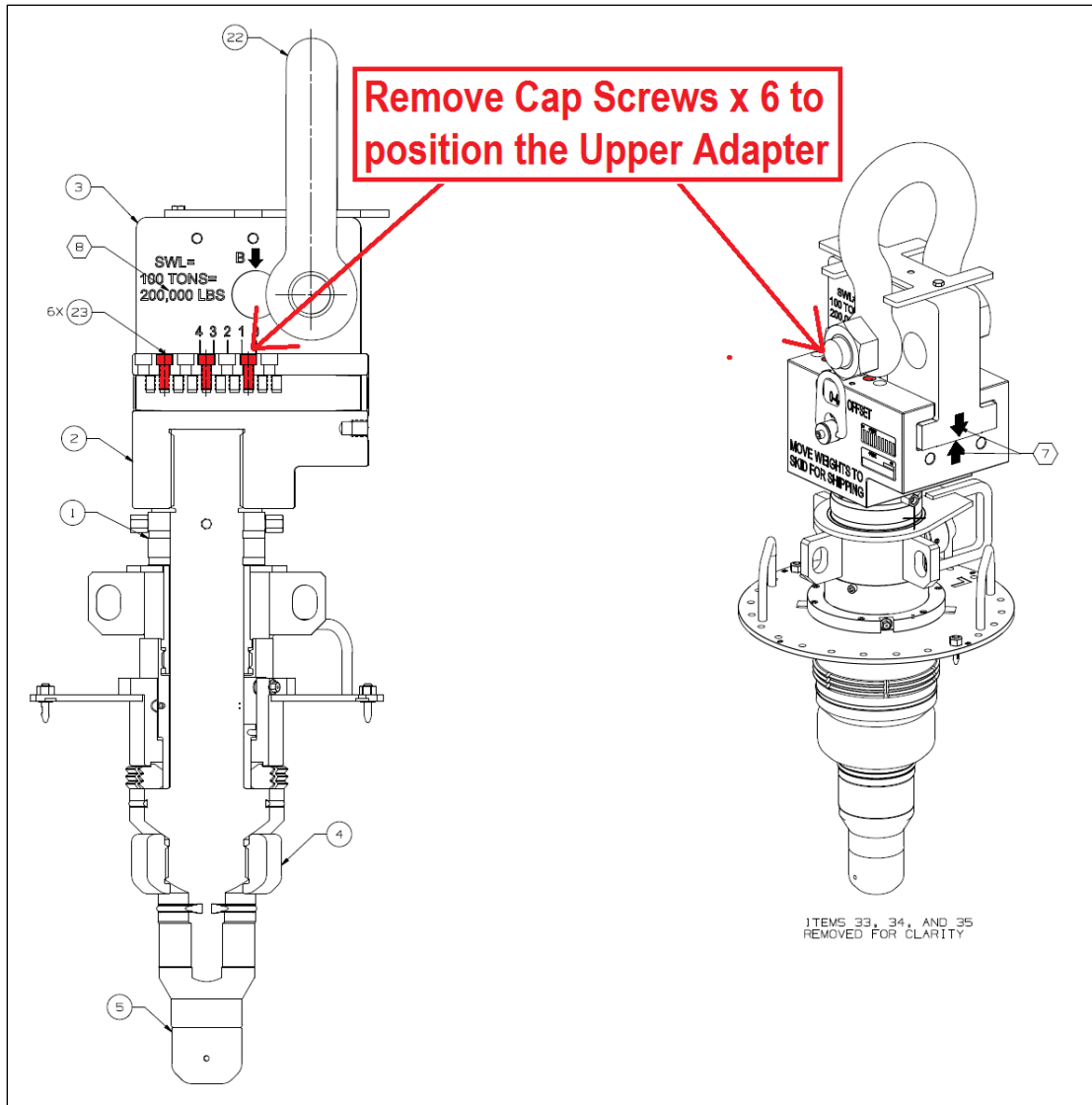


Figure 5: Linear Offset Cap Screw Location

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3. Slide the Upper Adapter (item 3) to the correct linear offset position. Reference the offset recorded in Section 5.1. Reference Figure 6 for offset positions.

NOTE: For 1/2-inch increments, align the scale marks half-way between the whole numbers.

- Use the scale on the "A" side for 5" to 7.5" linear offsets.
- Use the scale on the "B" side for 0" to 4.5" linear offsets.

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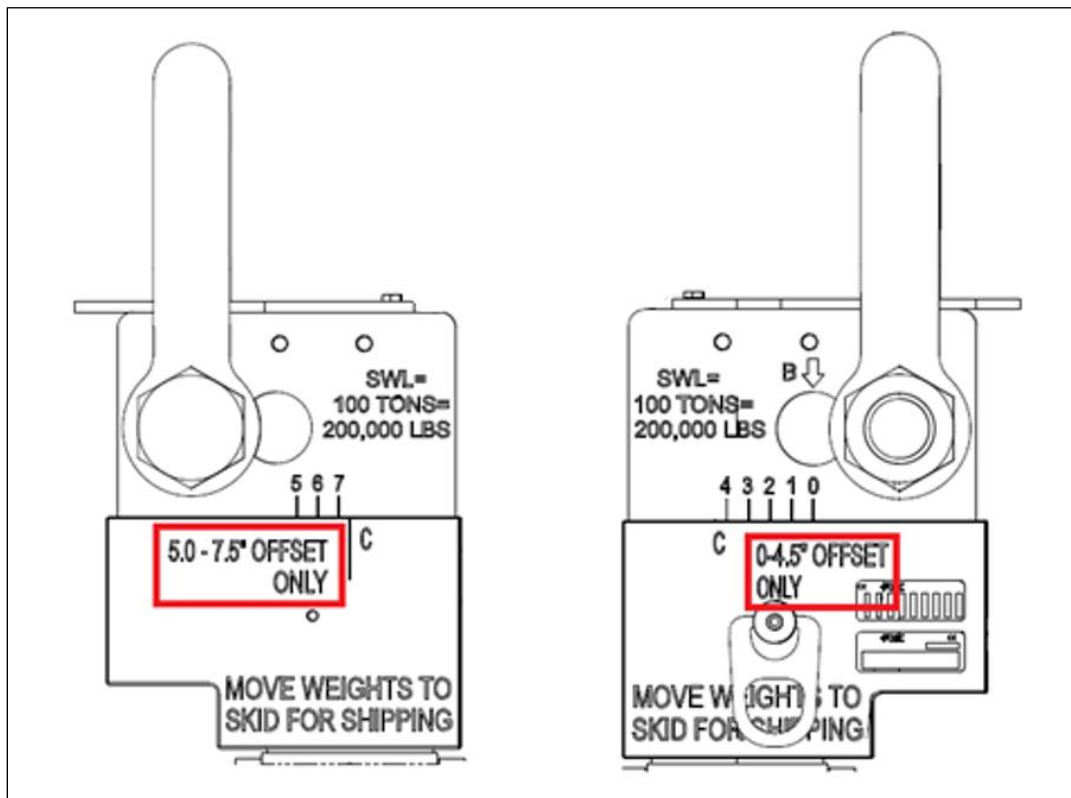


Figure 6: Offset Position

4. Verify that the alignment arrows, shown in red, line up with each other. If the arrows are on opposite sides of the tool then the Upper Adapter is not assembled correctly. See Figure 7.

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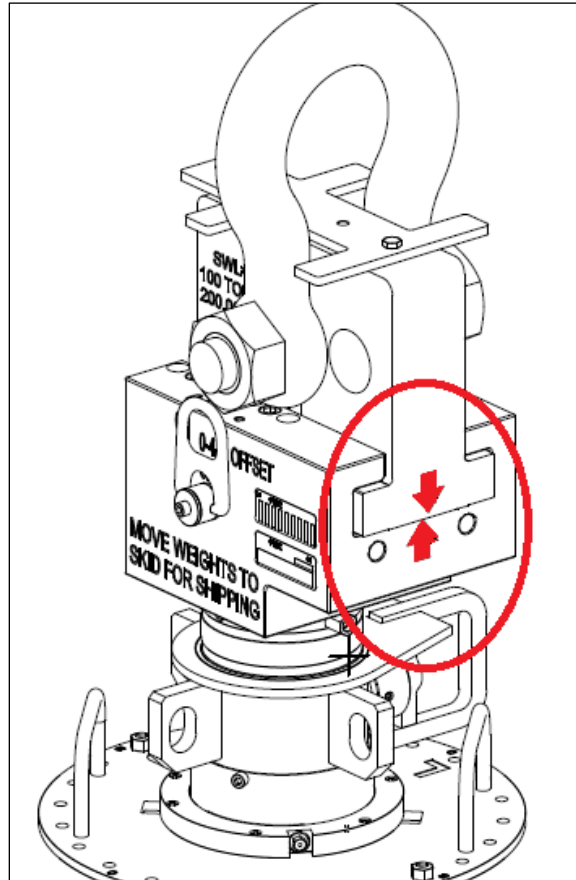
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Figure 7: Arrows Position

5. Reinstall the Qty. 6 Cap Screws (item 23) that were removed in step 1. These screws are used to lock the position of the Upper Adapter (item 3) with respect to the Lower Adapter (item 1). Torque screws hand tight (approx. 10-40 ft-lbs.). See Figure 8.

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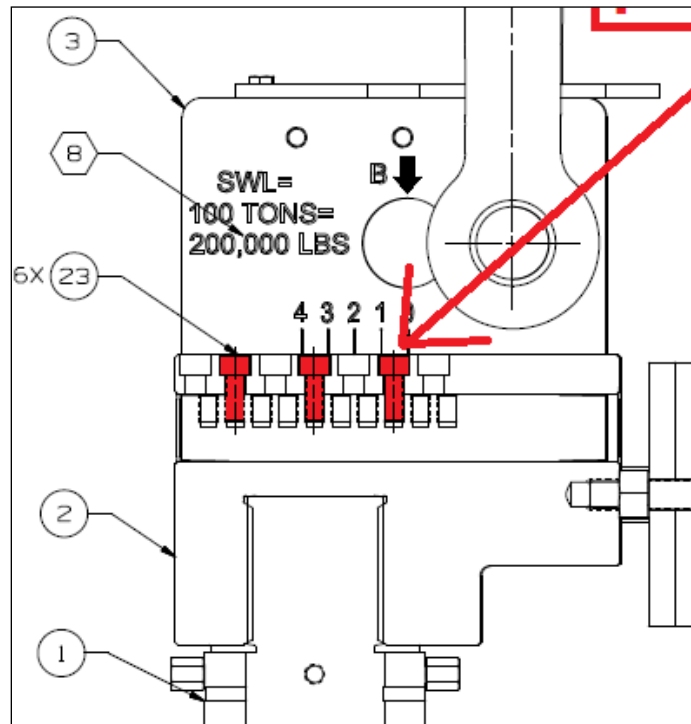
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Figure 8: Cap Screws



NOTE

A 0.5 inch offset only requires 4 screws. Bag and tag the other two cap screws while using a .5 inch offset, because the other two screws are used for every other offset.

5.2.2 Counterweight Installation

1. Reference the required linear offset from section 5.1.
 - If the linear offset is 3.5" or more, install Qty. 2 of the 21" long threaded rods until they bottom out. See Figure 9.
 - If the Linear offset is 3.0" or less, mark the remaining steps in this section "NA" and proceed to the next section labeled "Configure Angular Offset."

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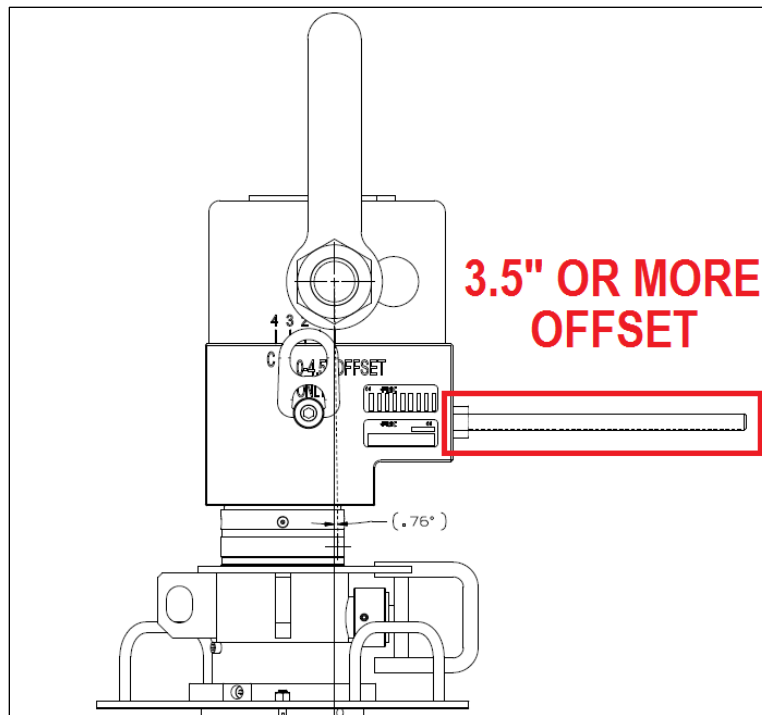
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Figure 9: Rods Installation

2. Measure the threaded rod protrusion and confirm that the measurement is LESS THAN 20.0 inches. See Figure 10.

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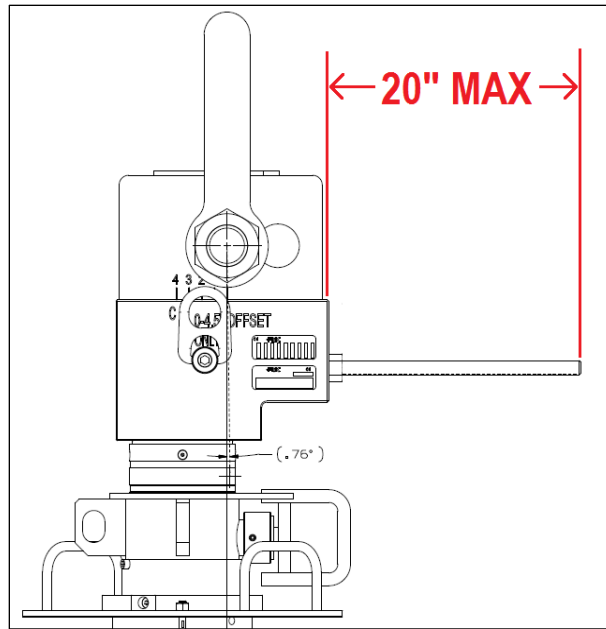


Figure 10: Rods Measurement

3. Tighten the 1.25" nuts against the Lower Adapter using approximately 50 to 100 ft-lbs. See Figure 11.

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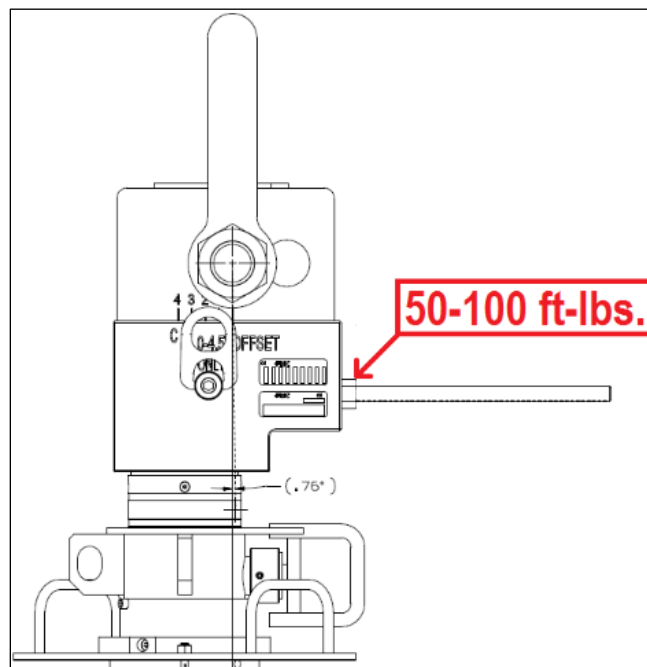


Figure 11: Rods Tighten

4. Reference the required linear offset from section 5.1 and install counterweights as needed:
 - For 0" to 3" offsets, counterweights are not required. Leave counterweights parked on shipping stand. See Figure 12.

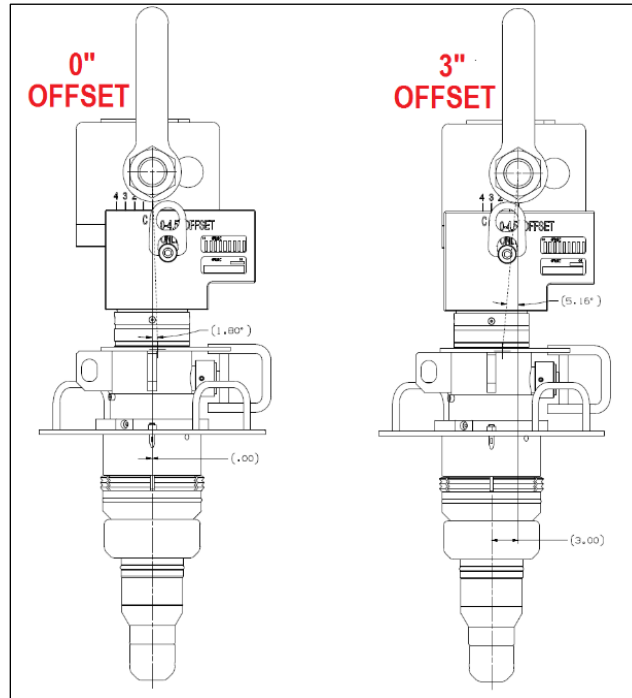


Figure 12: 0" to 3" offset

- For 3.5" to 4.5" offsets, install 8 counterweights and secure with Qty. 2, 1.25" nuts. Park remaining 8 counterweights on shipping stand. See Figure 13.

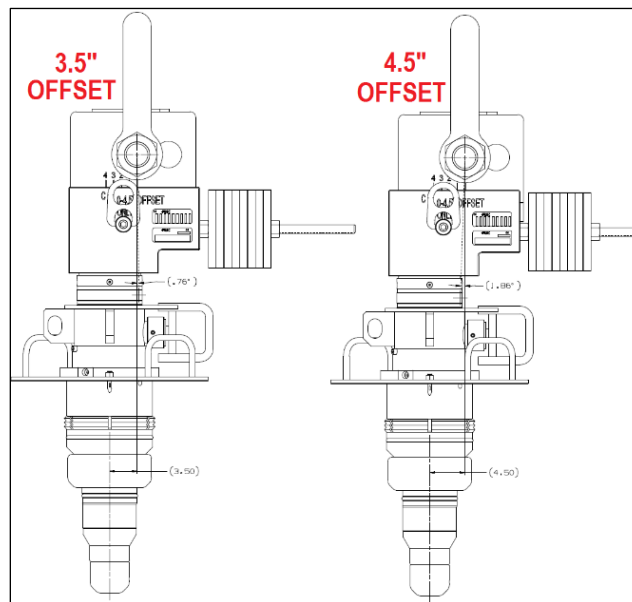


Figure 13: 3.5" to 4.5" offset

- For 5.0" to 7.5" offsets, install 16 counterweights and secure with Qty. 2, 1.25" nuts. See Figure 14.

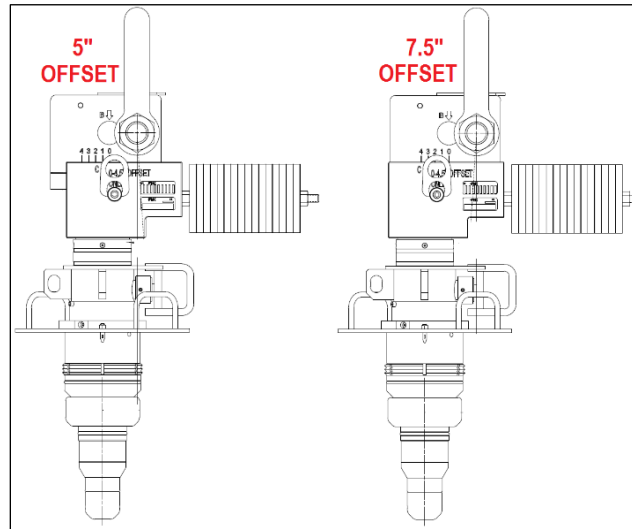


Figure 14: 5.0" to 7.5" offset

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DU100162251 shows the relationship between various COG offset settings and the quantity of counterweights corresponding to those settings.

NOTE

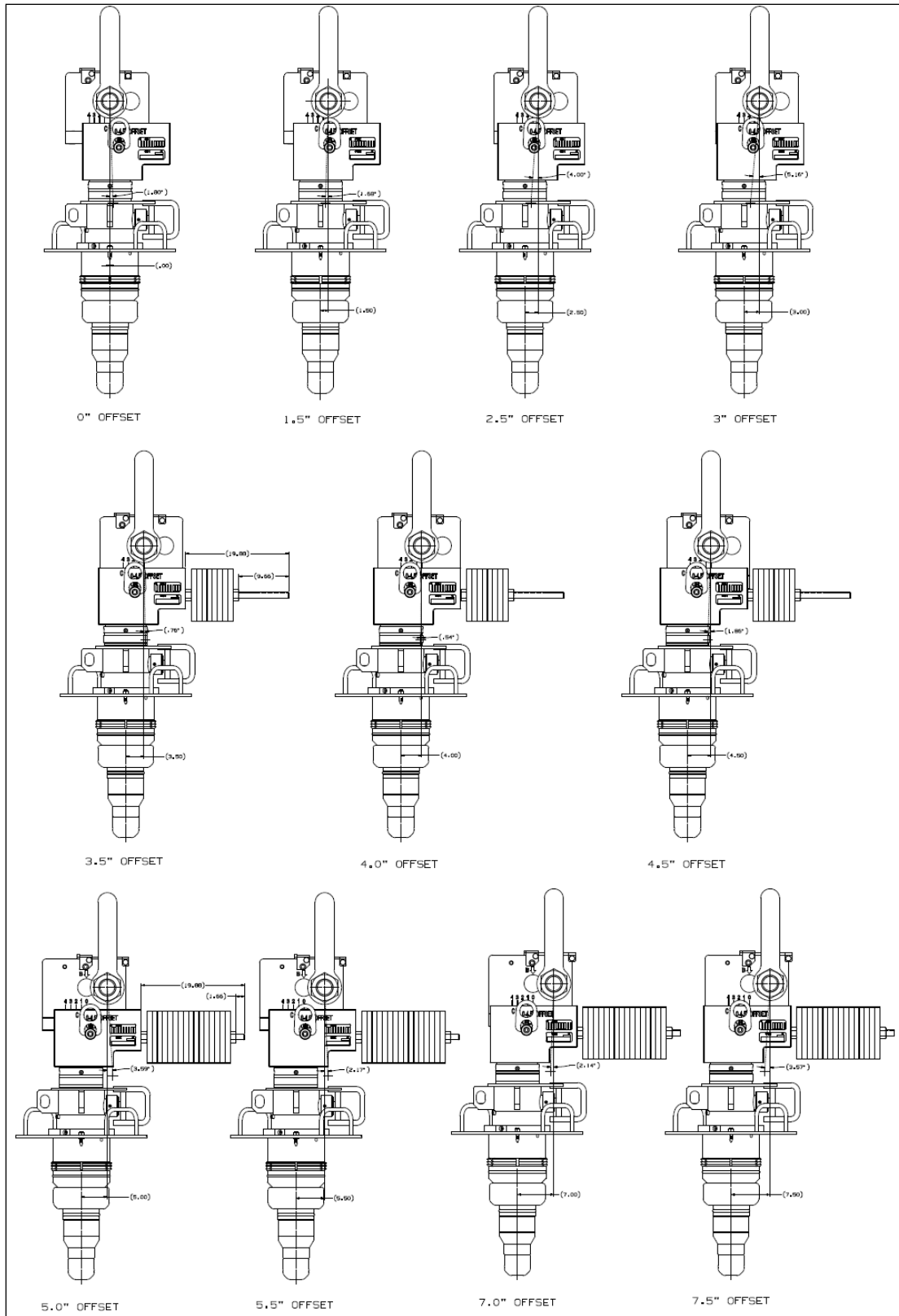


Figure 15: COG Offset Settings

5.3 Configure Angular Offset (if applicable)



NOTE

In this section, the Guide Pins on the Top Plate of the MTRT will be assembled to the position required for a given angular offset. This offset was previously determined in Section 5.1. The Guide pins are intended to interface with the handling holes on the left and right sides of the CVB, as shown below.

Also, a black stripe will be marked on the Top Plate using black electrical tape. This stripe is intended to match up with the front of the XT. The front of the XT is the side with the main ROV panel and Master Valve actuator.

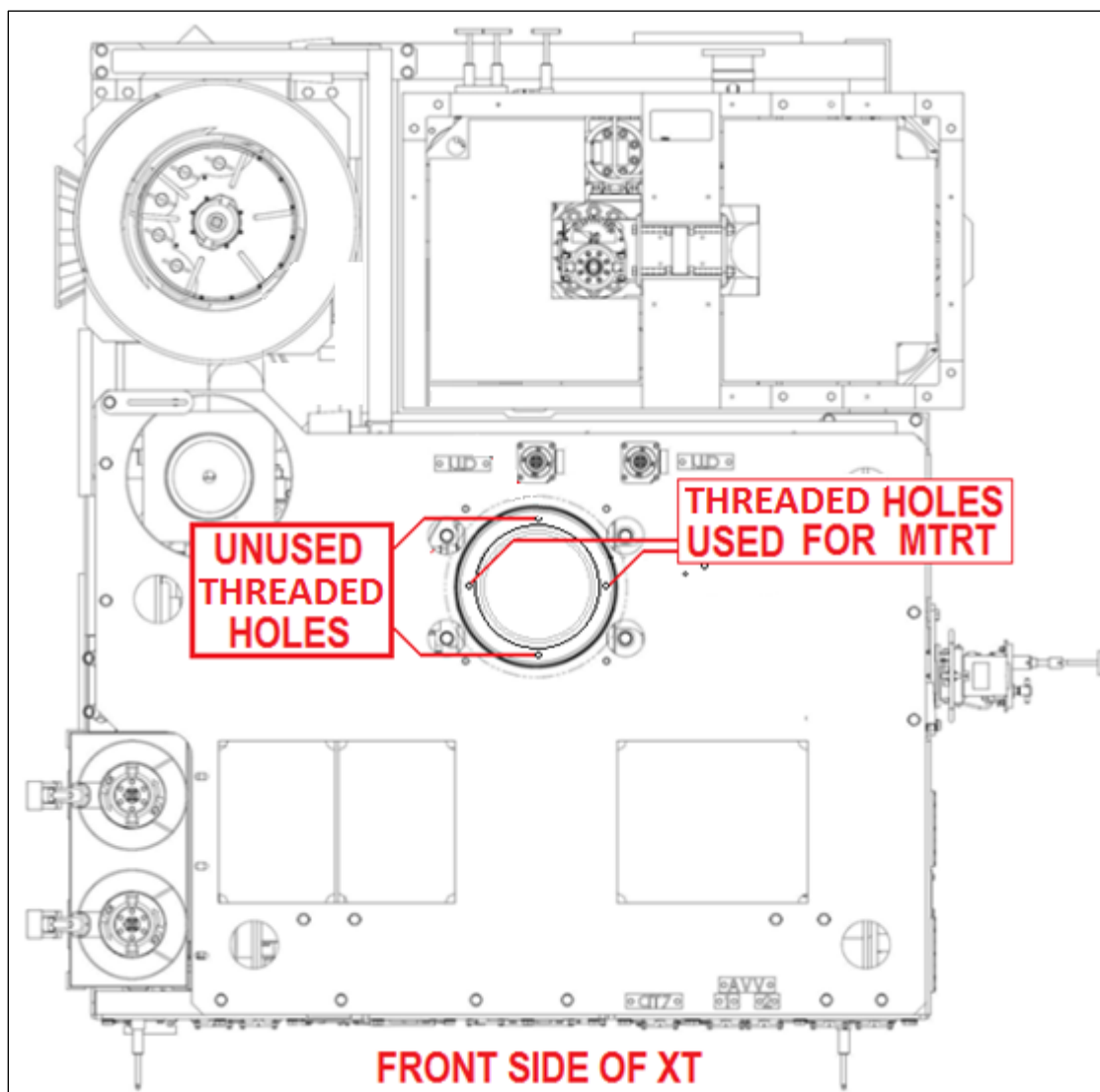


Figure 16: Guide Pins Position

Table 1: Angular Offset

Required Angular Offset	Guide Pin Position	Tape Position
Column 1	Column 2	Column 3
0	0	90 / white
15	15	105 / white
30	30	120 / white
45	45	135 / white
60	60	150 / white
75	75	165 / white
90	90	0 / yellow
105	105	15 / yellow
120	120	30 / yellow
135	135	45 / yellow
150	150	60 / yellow
165	165	75 / yellow
180	0	90 / yellow
195	15	105 / yellow
210	30	120 / yellow
225	45	135 / yellow
240	60	150 / yellow
255	75	165 / yellow
270	90	0 / white
285	105	15 / white
300	120	30 / white
315	135	45 / white
330	150	60 / white
345	165	75 / white

1. Record Project Specific Tape Position:

Tape Position: _____

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Date

2. See Table 1, Column 1, and find the row with the correct "Required Angular Offset." Circle this entire row.
 - NOTE: "Required Angular Offset" was obtained from Section 5.1. See Figure 17.

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Table 1: Angular Offset

Required Angular Offset	Guide Pin Position	Tape Position
Column 1	Column 2	Column 3
0	0	90 / white
15	15	105 / white
30	30	120 / white
45	45	135 / white
60	60	150 / white
75	75	165 / white
90	90	0 / yellow
105	105	15 / yellow
120	120	30 / yellow
135	135	45 / yellow
150	150	60 / yellow
165	165	75 / yellow
180	0	90 / yellow
195	15	105 / yellow
210	30	120 / yellow
225	45	135 / yellow
240	60	150 / yellow
255	75	165 / yellow
270	90	0 / white
285	105	15 / white
300	120	30 / white
315	135	45 / white
330	150	60 / white
345	165	75 / white

Figure 17: Angular Example

3. From Column 2, read the correct "Guide Pin Position." This number from Column 2 corresponds with two numbered positions on the Top Plate - these positions are where the Guide Pins are to be assembled. See Figure 18.

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Table 1: Angular Offset

Required Angular Offset	Guide Pin Position	Tape Position
Column 1	Column 2	Column 3
0	0	90 / white
15	15	105 / white
30	30	120 / white
45	45	135 / white
60	60	150 / white
75	75	165 / white
90	90	0 / yellow
105	105	15 / yellow
120	120	30 / yellow
135	135	45 / yellow
150	150	60 / yellow
165	165	75 / yellow
180		90 / yellow
195		105 / yellow
210		120 / yellow
225		135 / yellow
240	60	150 / yellow
255	75	165 / yellow
270	90	0 / white
285	105	15 / white
300	120	30 / white
315	135	45 / white
330	150	60 / white
345	165	75 / white

Figure 18: Linear Example

4. Move the two Guide Pins and two Hex Nuts to the Top Plate at the position read from Column 2. See Figure 19.

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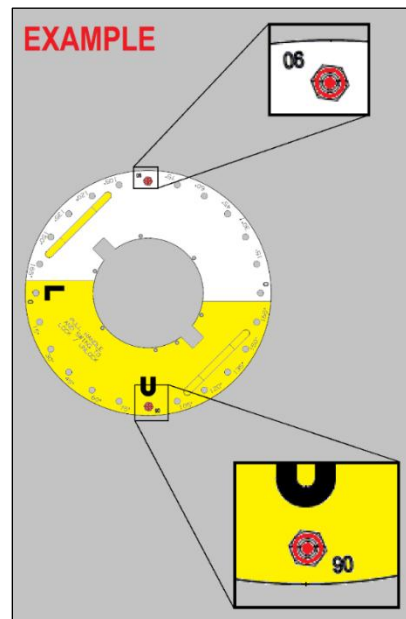


Figure 19: Pins and Nuts Position

5. From Column 3, read the "Tape Position" and use one piece of black electrical tape to mark the Top Plate at the position found in Column 3.
 - Only apply one piece of tape, and ensure that the tape is on the correct colored side of the Top Plate (white or yellow).
 - Apply the tape over the hole in the Top Plate and orient the tape towards the center of the MTRT, as shown on Figure 20.

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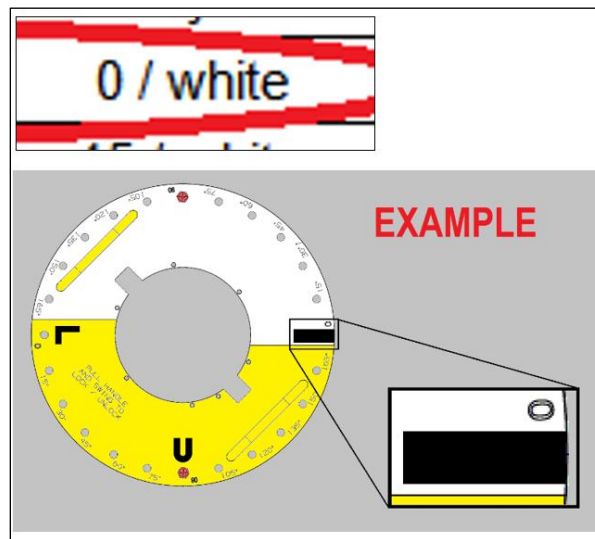
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Figure 20: Tape Position

5.4 Inspect MTRT

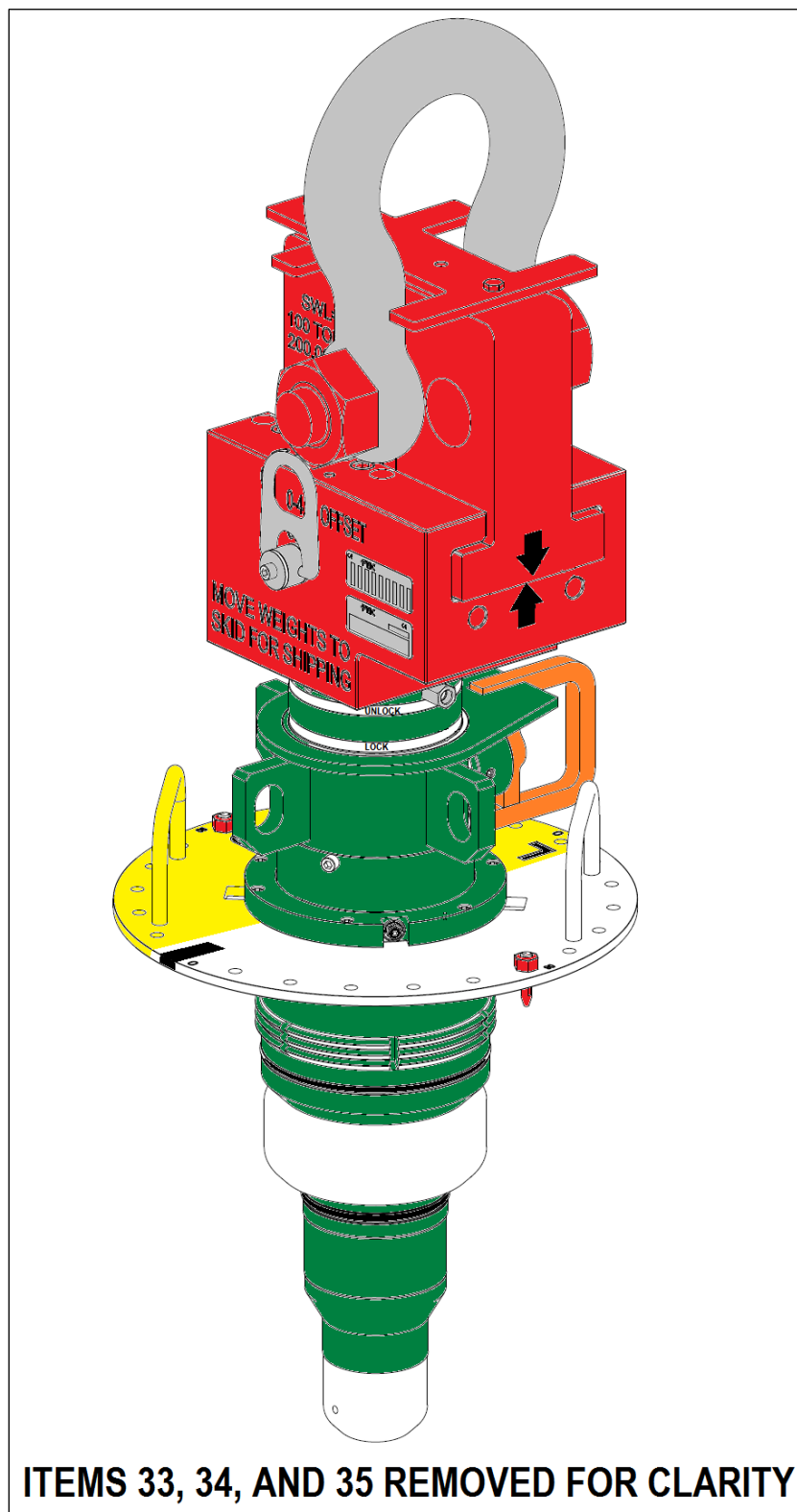


Figure 21: MTRT As-Built

1. Compare the MTRT to the part report and Figure 21 above to ensure that the tool has not been modified from its original as-manufactured form.

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2. Inspect the MTRT for any signs of wear & tear or loose parts.

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3. Ensure all of the cap screws that fasten the Upper Adapter to the Lower Adapter are bottomed out. Reference Table 2.

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DateTable 2: Cap Screws Recommended Torque (ref. [SWI20014633](#))

Cap Screw Size	Quantity	Torque Valve (ft-lb)
0.375 - 16UNC	7	10
0.500 - 13UNC	2	25
0.750 - 10UNC	6	90

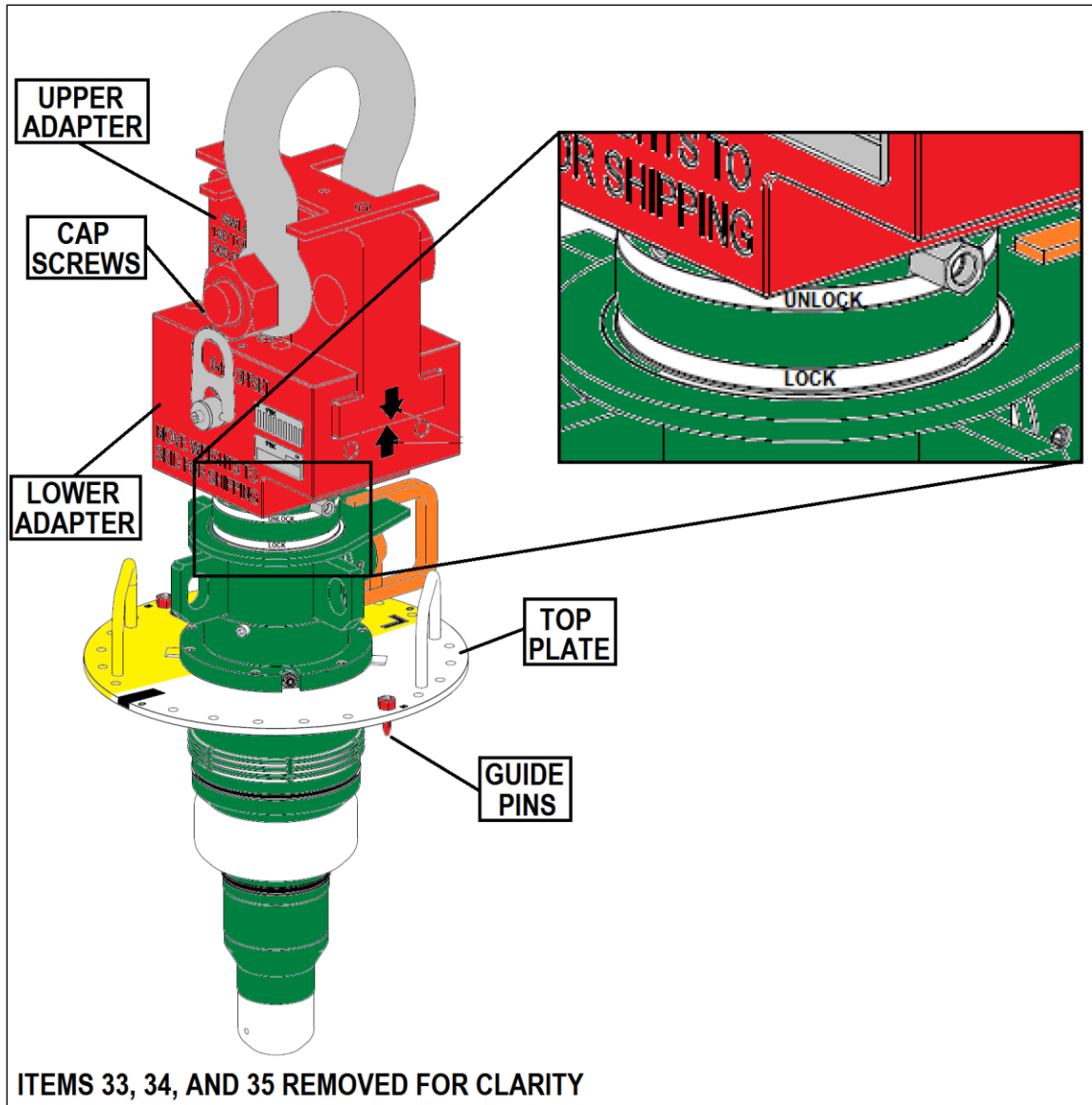


Figure 22: MTRT Feature Identification

4. While the MTRT is still installed in the Shipping Stand, Inspect the following components:
 - Main Body – Verify the Lock and Unlock grooves are painted and visible.
 - Top Plate – Verify the entire plate is flat and shows no signs of major damage. Verify the markings "L" ,"U" , and the black orientation tape strip can all be clearly seen by an ROV
 - Ensure the Hex Nuts on the Guide Pins are secure.

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5. Unlock the MTRT by pulling the ROV Handle away from the Main Body by about $\frac{1}{2}$ " and then rotating the ROV Handle around the Main Body by $\frac{1}{4}$ turn in the CCW direction (right) until it is automatically retained in place in the fully "Unlock" position. (See Figure 23).

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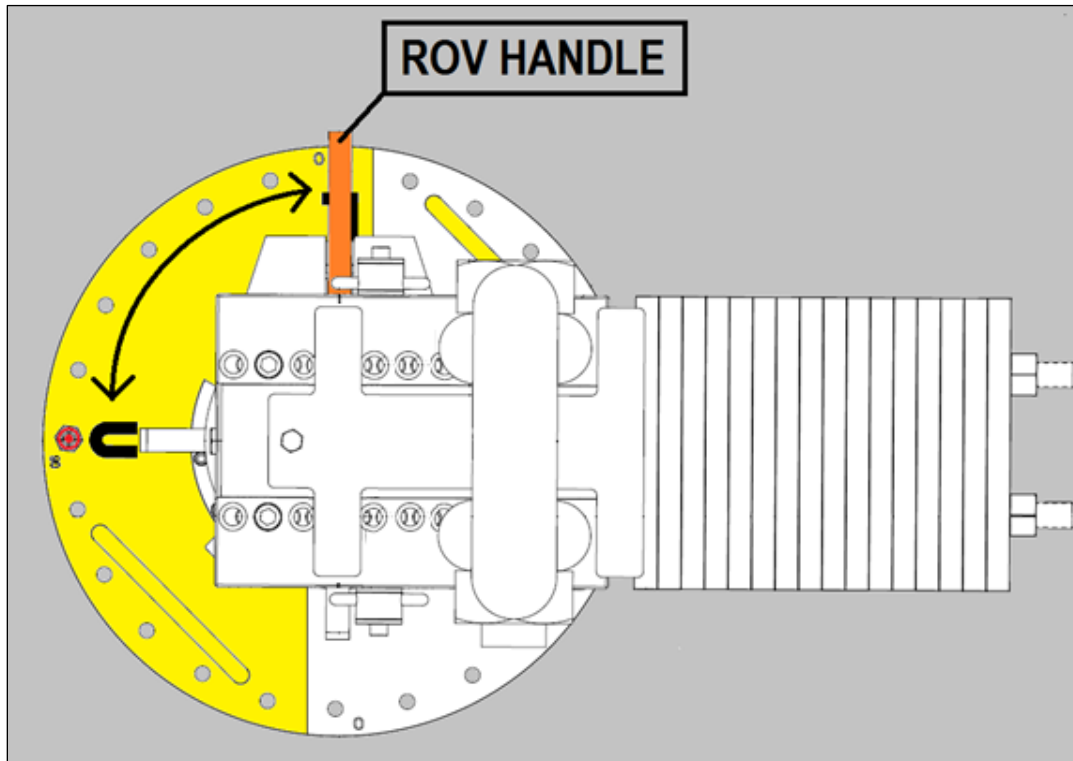


Figure 23: Rotation of ROV Handle from Lock to Unlock

6. ROV Handle –Verify the Handle functions to Lock and Unlock and there is no damage or loose components. Verify the internal spring holds the Handle in the full Locked and unlocked positions by trying to move from lock or unlock without pulling the handle.

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Date



If 10K XT will be involved in the lifting process it is essential to ensure that the sling is not twisted.

CAUTION

7. Lift MTRT out of the Shipping Stand. Store Shipping Stand in a secure area nearby, preferably on a soft surface/ wooden pallet. It will be needed immediately after retrieving the MTRT.

Initial

Date

8. Visually inspect the Locking Segments and verify there is no debris on the Segments. If debris is noticed, wash with fresh water. Also inspect to ensure that the locking segments are evenly spaced with respect to each other while the MTRT is in the unlocked position.

Initial

Date

9. Function the ROV Handle 90 degrees ($\frac{1}{4}$ turn) between Lock and Unlock several times. Verify the ROV Handle and locking mechanism operates smoothly and properly:
 - Verify the ROV Handle can be pulled out and swung in between the Lock and Unlock positions by hand.
 - Verify the top of the ROV Handle Guide lines up with the bottom of the "Lock" groove on the Main Body when Locked and with the bottom of the "Unlock" groove when Unlocked. (See Figure 23)
 - Verify the markings on the Top Plate line up correctly with the ROV Handle at full Lock and Unlock. Verify the "L" is on the Lock side and the "U" is on the Unlock side.
 - When in the Lock position, verify the Lock Segments are fully engaged and forced out by the Lock Mandrel.
 - When in the Unlock position, verify the Lock Segments are completely disengaged and do not contact the Lock Mandrel. Verify the Lock Segments do not protrude past the larger OD of the Main Body when pushed back against the smaller OD of the Main Body.
 - With the ROV Handle in the unlocked position, verify the Lock Segments cannot fall out of the MTRT. They should be held in by the Segment

Retainer. Rotate and tilt the Segments forward, backward, and side to side.

Initial

Date


NOTE

The MTRT should have no external seals when used to install or remove a XT Subsea. Leaving the seal in place may cause drag or hydraulic lock that could prevent complete tool engagement into the CVB.

10. Ensure the two S-seals on the OD of the Main Body are NOT installed. If they are installed, remove and store them.

Initial

Date

11. Inspect the following components (Reference Figure 1 & Sub-Assembly DWG [DU100239949](#)):

- Locking Segments – Verify the Locking Profile is smooth without any sharp edges. Verify the Segments show no signs of yielding. Ensure there is no dirt or debris between or behind the Locking Segments.
- Bumper & Stab Nose – Verify the Stab Nose and Bumper are secure and tight. Verify there is no major damage to the parts.
- Main Body - Verify there is no damage or sharp edges on the outer profile. The OD must be a smooth and continuous surface to avoid damaging the XT internal components and seal surfaces.

Initial

Date


For installation of a XT, proceed to Section 7.0. For retrieval of a XT, proceed to Section 8.0.

NOTE

6.0 Handling MTRT Without XT



If using swivel hoist rings to lift MTRT Assembly, ensure no Counterweights are installed to MTRT. If Counterweights are installed, lift using 3"-120 Ton Shackle.

CAUTION Refer to [THI10064637](#) for more details.



NOTE

MTRT shall be lifted using the 3" 120 Ton shackle or Swivel Hoist Rings (2x) may be used for lifting and handling MTRT assembly (without XT) only for MTRT offset settings of 3.0" or less. MTRT offset settings of 3.0" or less do not require the use of any counterweights.

When handling the MTRT assembly with no counterweights or half the number of counterweights, park the remaining counterweights on the shipping stand. Refer to [DU100162251](#) for relationship between number of counterweights to be used and corresponding MTRT offset settings.

1. The area where the lifting operations take place must be clear of obstructions.
2. During lifting operations personnel in the area must be warned to stand clear.

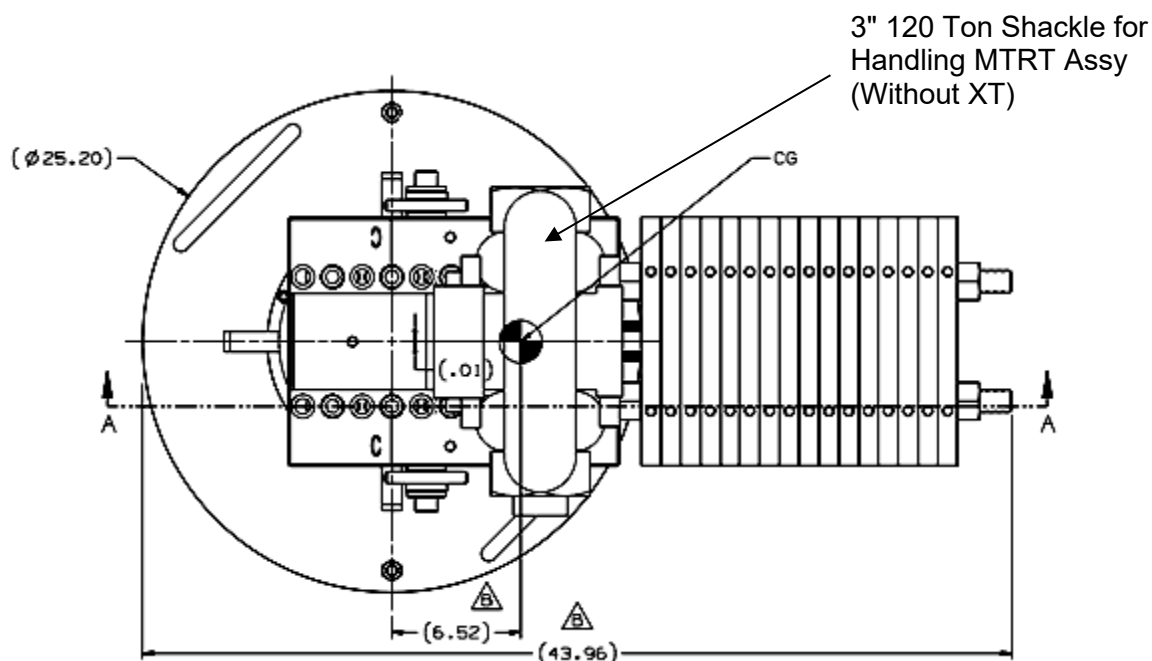


Figure 24: MTRT Top View Dimensions and CoG (MTRT Shown with 7.5 in Offset for Reference)



Figure 24 shows the MTRT COG with respect to the geometric center of the assembly at a Shackle Radial Offset of 7.5", for reference.

NOTE

6.1 Lifting Instructions With 120 Ton Shackle

1. Follow instructions for handling MTRT per [THI10064637](#) (Section 2.3.1) and attach signed copy to the appendix.

Initial

Date

6.2 Lifting Instructions With Swivel Hoist Ring (Only For Offset Settings Up To 3.0", Without Counterweights)

1. Follow instructions for handling MTRT per [THI10064637](#) (Section 2.3.2) and attach signed copy to the appendix.

Initial

Date

7.0 Running XT with MTRT



Section 7.0 is only to be performed if the MTRT is to be used to Install a XT. See Section 8.0 for Retrieving a XT.

NOTE

7.1 Dry-Run the MTRT



This section should be performed prior to installation of the XT. The intent is to verify that the MTRT will perform as intended when operated Subsea. This will include locking the tool to the XT and preparing to run subsea, then removing the Tool from the XT as the ROV would subsea.

NOTE

1. Section 5.0 of this procedure has been completed.

Initial

Date

2. Verify the tool is in the MTRT configuration (remove both external S-seals) according to the tool Sub-Assembly DWG ([DU100239949](#), minus the s-seals).

Initial

Date

3. On the deck, Install and Lock the MTRT according to Section 7.2 below.

Initial

Date

4. On the deck, Unlock and Remove the MTRT from the XT as the ROV would, according to Section 7.3.

Initial

Date

7.2 Install the MTRT to the XT

1. Unlock the MTRT by pulling the ROV Handle away from the Main Body by about ½" and then rotating the ROV Handle around the Main Body by ¼ turn in the CCW direction, or right as seen from the side. until it is automatically retained in place in the fully "Unlocked" position. See Figure 23.

Initial

Date

2. Using the 3" 120 Ton shackle, Install the MTRT into the XT, taking care not to bump XT seal areas, XT locking profile, or MTRT Lock Segments. The stab nose and bumper will minimize any risks of damaging the seal areas. The Top Plate should be about 0.17" from the top of the XT hub when the MTRT is fully landed. See Figure 25.

Initial

Date

3. Record the land-out measurement if the XT is topside, or visually verify land-out if the XT is subsea.

Topside: LAND-OUT MEASUREMENT = _____ Inches

(MUST be LESS THAN 1/4 inch)

Subsea: LAND-OUT must be verified by confirming that the gap below the Top Plate is less than half of the thickness of the Top Plate (LESS THAN 1/4 inch).

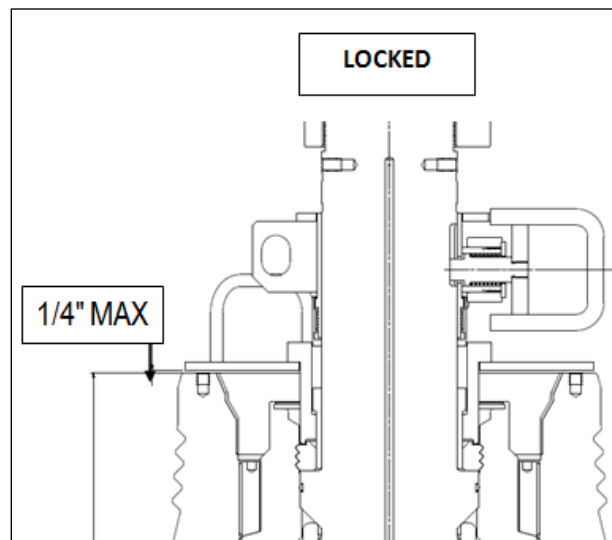


Figure 25: MTRT Land-Out

Initial

Date

4. If Applicable, ensure that the Guide Pins (2X) line up with the correct handling holes on the XT valve block to ensure that the COG of the XT will line up with the radial and angular offset of the 120 ton shackle. NOTE: The black tape on the Top Plate of the MTRT should align with the front / master valve side of the XT.

Initial

Date

5. Lock the MTRT / XTHT by pulling the ROV Handle away from the Main Body by about ½" and then rotating the ROV Handle around the Main Body by ¼ turn CW until it is in the fully "Locked" position.

Verify the top of the ROV Handle Guide lines up with the bottom of the "LOCK" groove on the Main Body. The "LOCK" groove must be fully visible. See Figure 26.

Initial

Date

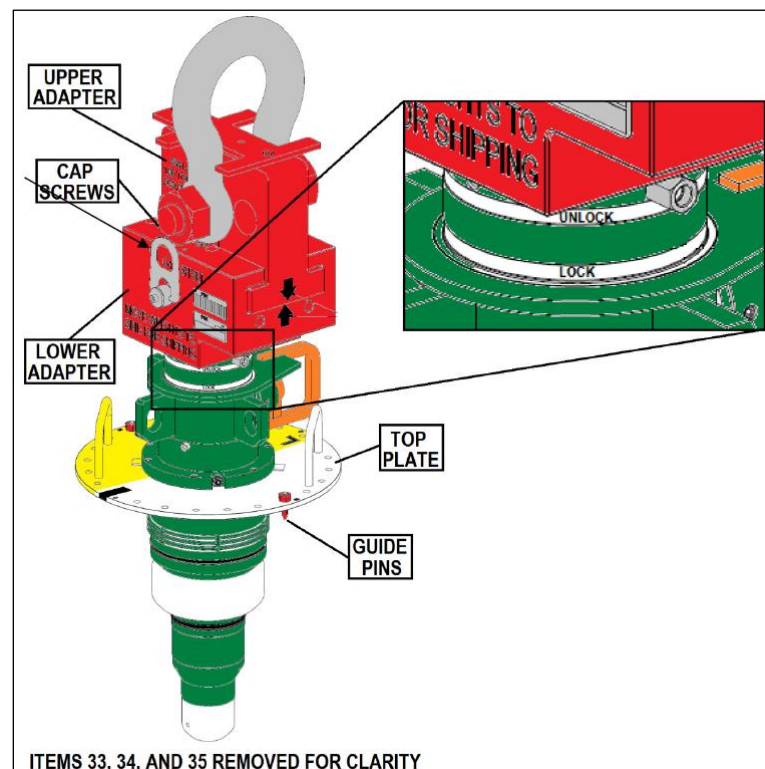


Figure 26: Front View of MTRT in Locked Position Showing Lock Groove on Main Body



If 10K XT will be involved in the lifting process mark position of MTRT in respect to XT by drawing line between them.

NOTE

6. Run the XT subsea per the XT Running Procedure(s).

Initial

Date

7.3 Recover the MTRT

7.3.1 Normal Release



There might be some displacement of water while functioning the tool, causing the ROV Handle to move slowly.

NOTE

The ROV will not be able to unlock the MTRT if there is tension in the rigging setup.

1. Slack off slightly on the installation wireline or drillpipe. It is desirable that the wireline is near vertical during retrieval of the MTRT ($\pm 5^\circ$ max).

Initial

Date



WARNING

If swivel is not used in the rigging directly above the MTRT, it is extremely important that selected rigging configuration be oriented directly above the lift point on the MTRT and within alignment to less than 5 degrees. Ensure to release any torsion build up in the rigging configuration prior to performing any lift. Ensure torque build-up due to rigging or external effects does not exceed 12,500 ft-lbs.

Failure to follow these guidelines can result in MTRT being subjected to excessive torque.

2. Have the ROV hold onto one of the two grab bars on the Top Plate and pull the ROV Handle out from the Main Body by about ½" and move the handle only a few inches to the right (CCW) and release the handle. The ROV can then push the ROV Handle around the Main Body to complete the full ¼ turn in the CCW direction until it is automatically retained in place in the fully "Unlocked" position. As the handle rotates around the tool, it will also translate vertically upwards.
 - Have the ROV verify the top of the ROV Handle Guide lines up with the bottom of the "Unlock" groove on the Main Body.
 - Have the ROV verify the ROV Handle lines up correctly with the markings on the Top Plate on the "U", or Unlock, side.

Initial

Date

SECONDARY RELEASE:

If the ROV cannot unlock the MTRT due to unforeseen circumstances, there is an option for Secondary Release. See Section 7.3.2.



NOTE

Before resorting to secondary release, have the ROV attempt to swing the ROV Handle in the CCW direction (after pulling the handle out at least ½") with as much force as the ROV operator is comfortable with. The MTRT can handle a maximum of 1000 lbs applied to the ROV Handle in the direction tangent to the axis of the ROV Handle, as shown in Figure 27.

If the ROV cannot unlock the MTRT, proceed to 7.3.2 for Secondary release.

3. Retrieve the MTRT back to the surface using the center shackle and wireline.

Initial

Date

4. Inspect and wash the tool thoroughly. Install and lock the MTRT into the Shipping Stand.

Initial

Date

7.3.2 Secondary Release

This section for Secondary Release should only be used in the event that the ROV cannot rotate the ROV Handle on the MTRT to Unlock the tool.



The Maximum Over pull Capacity of the ROV Handle Guide is 22.0 kips.

CAUTION Secondary Release involves shearing two internal pivot pins on the MTRT. The tool must then be dis-assembled on the surface to replace the two pivot pins so the MTRT can function again.

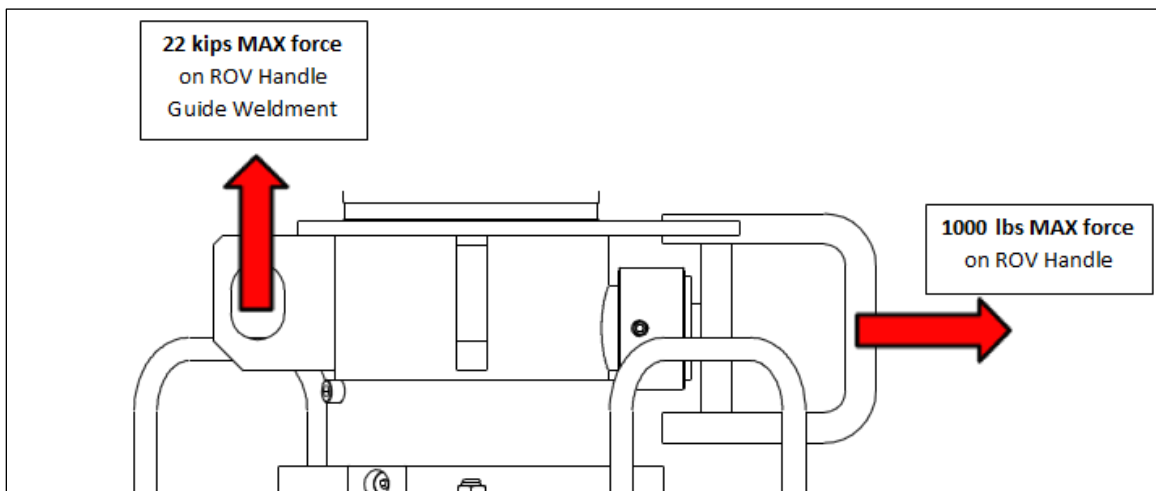


Figure 27: Maximum Force Limits on ROV Handle and ROV Handle Guide Weldment

1. The ROV is to attach the Wireline or a Rig Tugger Line to the padeye on the ROV Handle Guide Weldment located every 90 degrees from the ROV Handle. See Figure 1 for padeye orientation on the ROV Handle Guide Weldment. The interface is designed for a 7 ton Crosby Hook.

Initial

Date

2. The wireline or tugger line can then overpull with 10.0 kips to shear the two pivot pins, see Figure 29, located internal to the MTRT. (Maximum capacity of the ROV Handle Guide is 22 kips as seen in Figure 27).

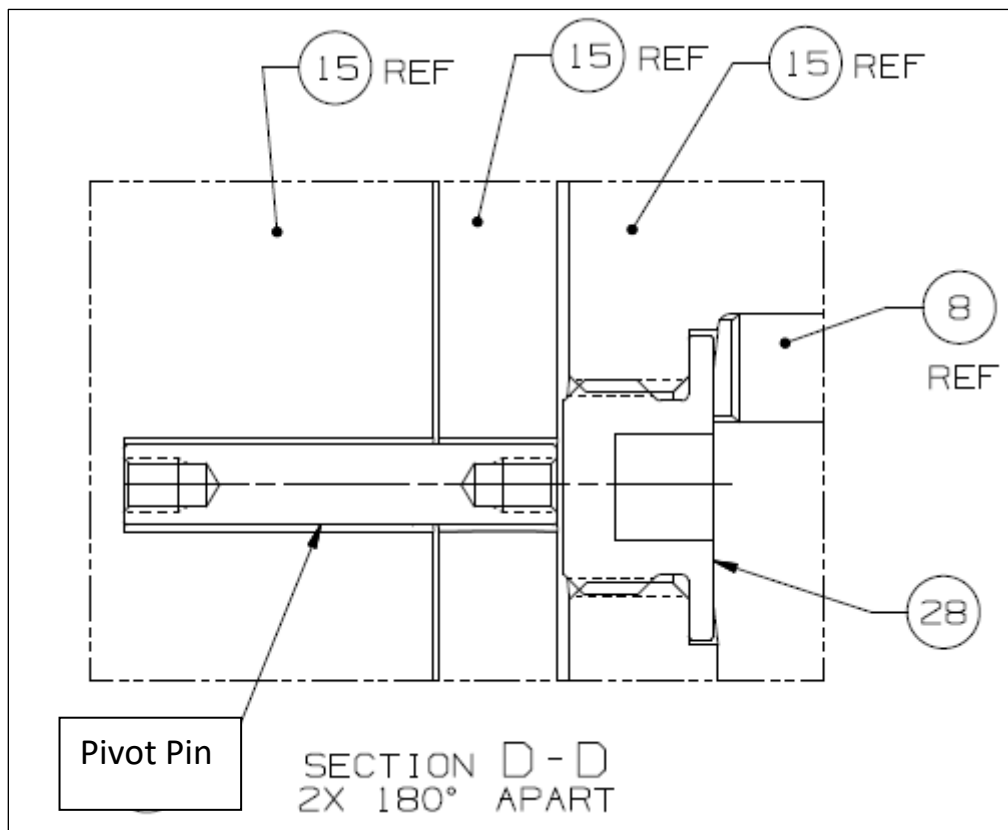


Figure 29: Location of Pivot Pin

Initial

Date

3. Verify the installation wireline is near vertical to prevent the MTRT from swinging excessively once removed from the Wellhead / Tubing Head. Retrieve the MTRT using the center shackle and wireline.

Initial

Date

4. Inspect and wash the tool thoroughly. Install and lock the MTRT into the Shipping Stand. The ROV Handle Guide, and the Lock Mandrel will fall automatically, once the pivot pins have been sheared. See Note below for further instructions.

Initial

Date

CAUTION

The MTRT must be installed into the Shipping Stand ([P1000170649](#)) anytime it is not Locked to the XT.

Do not lay down or bump the MTRT on the Top Plate or ROV Handle. Damage to these components may not allow the MTRT to install properly to the XT.


NOTE

Send the MTRT back to shore for replacement of the Pivot Pins and inspection, or if replacement pins are available, the pins can be replaced using the Sub-Assembly DWG [DU100239949](#) and [ASY10131969](#) (Only Assembly required).

8.0 Retrieving XT with MTRT



NOTE

Section 8.0 is only to be performed if MTRT is to be used to retrieve a XT. If at any time, the MTRT Locks accidentally, when the ROV Handle rotates into the Lock position, ROV can unlock the tool subsea. This operation may also be performed while the MTRT is partially inserted into the XT. There might be some displacement of water while functioning the tool, causing the ROV Handle to move slowly.

8.1 Install the MTRT to the XT and Retrieve



NOTE

Confirm that the heave compensation is adequate before having the ROV approach the MTRT. Excessive motion can result in damage to the MTRT, EHXT, or ROV. This is a particular concern when using an HCLS deployment method since the buoyancy modules need to be fully deployed to a sufficient depth to provide adequate heave compensation.

1. Section 5.0 of this procedure has been completed.

Initial

Date

2. Carefully lift the MTRT from the deck ensuring that it does not drag on the deck and become damaged. Quickly deploy the MTRT through the splash zone and lower it to roughly 50 feet above the tree.

Initial

Date

3. Ensure the TA cap or protective cover is removed from the tree while deploying the MTRT. Have the ROV fly to the MTRT location to help guide it down toward the bore of the tree.

Initial

Date

4. Have the ROV verify the MTRT is still fully unlocked, see Step 2 of Section 7.3 for verification. Lower the MTRT into the XT. The stab nose and bumper will minimize any risks of damaging the seal areas.

Initial

Date

5. Continue to slowly lower the MTRT into the top of the tree. Once landed, the Top Plate should be an average of about 0.17" from the top of the XT hub when the MTRT is fully landed. See Figure 25.

Initial

Date

6. If applicable, ensure that the Guide Pins (2X) line up with the correct handling holes on the valve XT block to ensure that the COG of the XT will line up with the radial and angular offset of the 3", 120 ton shackle.

Initial

Date

7. It is desirable that the installation wireline is near vertical ($\pm 5^\circ$ max) during Locking of the tool. Too much slack may pull the MTRT to one side and might cause Locking to be difficult for the ROV.

Initial

Date

8. Have the ROV hold onto one of the two grab bars on the Top Plate and pull the ROV Handle out from the Main Body by about $\frac{1}{2}$ " and move the handle only a few inches to the left (CW) and release the handle. The ROV can then push the ROV Handle around the Main Body to complete the full $\frac{1}{4}$ turn in the CW direction until it is automatically retained in place in the fully "Locked" position. As the handle rotates around the tool, it will also translate vertically down.
 - Verify the top of the ROV Handle Guide lines up with the bottom of the "Lock" groove on the Main Body. See Figure 26.
 - Verify the ROV Handle lines up correctly with the markings on the Top Plate on the "L", or Lock, side. See Figure 23.

Initial

Date


WARNING

If swivel is not used in the rigging directly above the MTRT, it is extremely important that selected rigging configuration be oriented directly above the lift point on the MTRT and within alignment to less than 5 degrees. Ensure to release any torsion build up in the rigging configuration prior to performing any lift. Ensure torque build-up due to rigging or external effects does not exceed 12,500 ft-lbs. Failure to follow these guidelines can result in MTRT being subjected to excessive torque.

9. Verify the installation wireline is near vertical to prevent the XT from swinging excessively once removed from the Wellhead / Tubing Head. Retrieve the XT using the center shackle and wireline.

Initial

Date

CAUTION

Ensure that the right sling size (right capacity) is used in handling the MTRT using drillpipe as listen under Section 4.3.2.

10. Once the XT is located on deck, the MTRT can be left in the XT for handling of the XT, or it can be installed into the Shipping Stand for storage.

Initial

Date

CAUTION

The MTRT must be installed into the Shipping Stand ([P1000019868](#)) immediately upon removing from the XT. Do not lay down or bump the MTRT on the Top Plate or ROV Handle Weldment. Damage to these components may not allow the MTRT to fully install to the XT.

9.0 Appendix

9.1 Top Plate Angular Orientation Examples

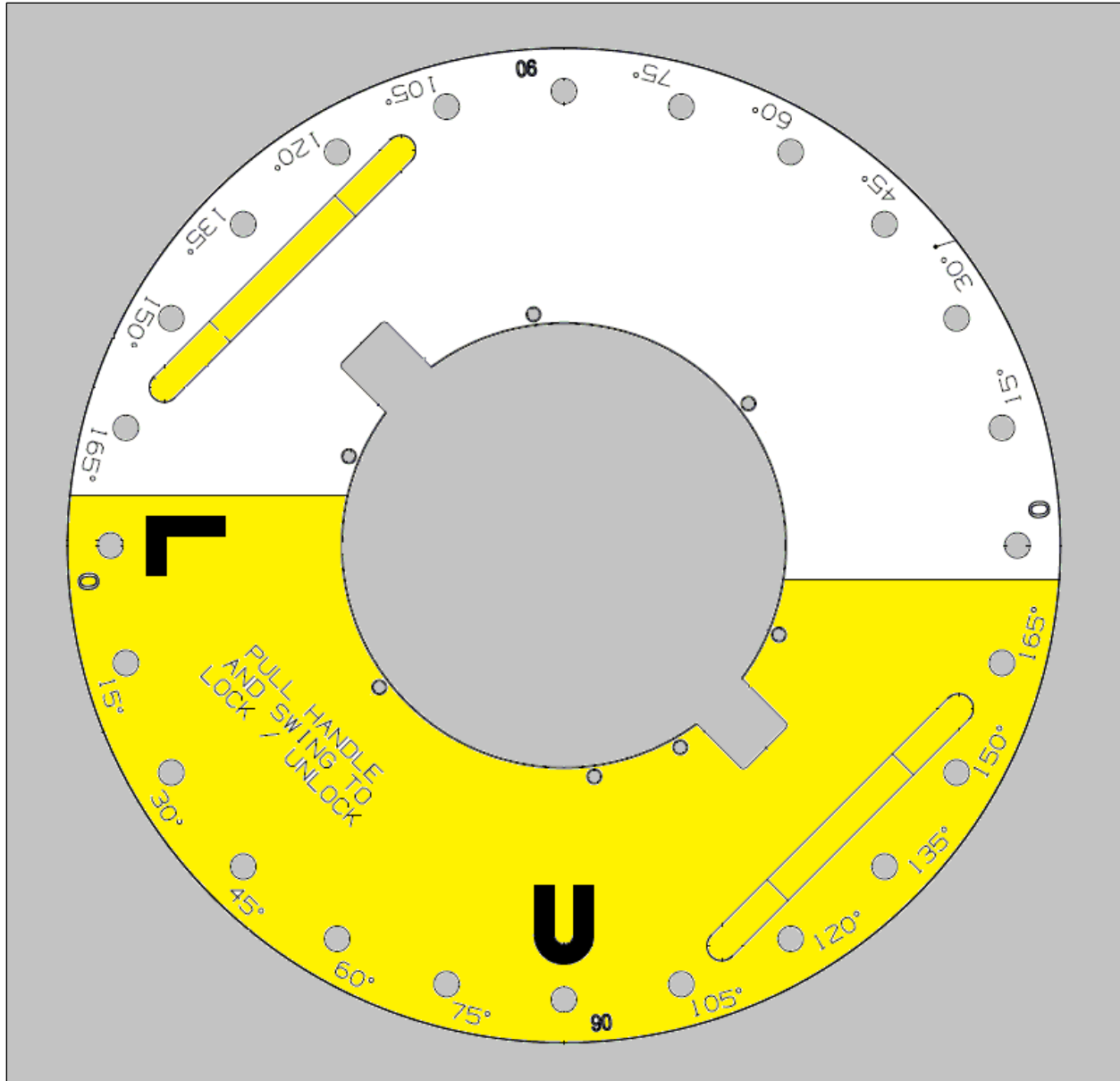


Figure 30: Dowel Pin Holes in Main Body to Set Angular Offset

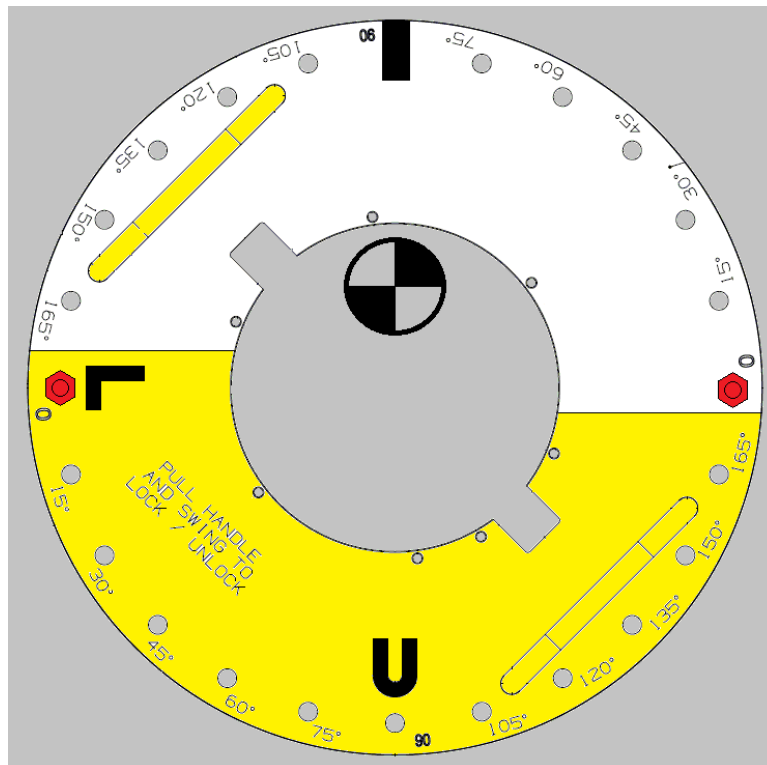
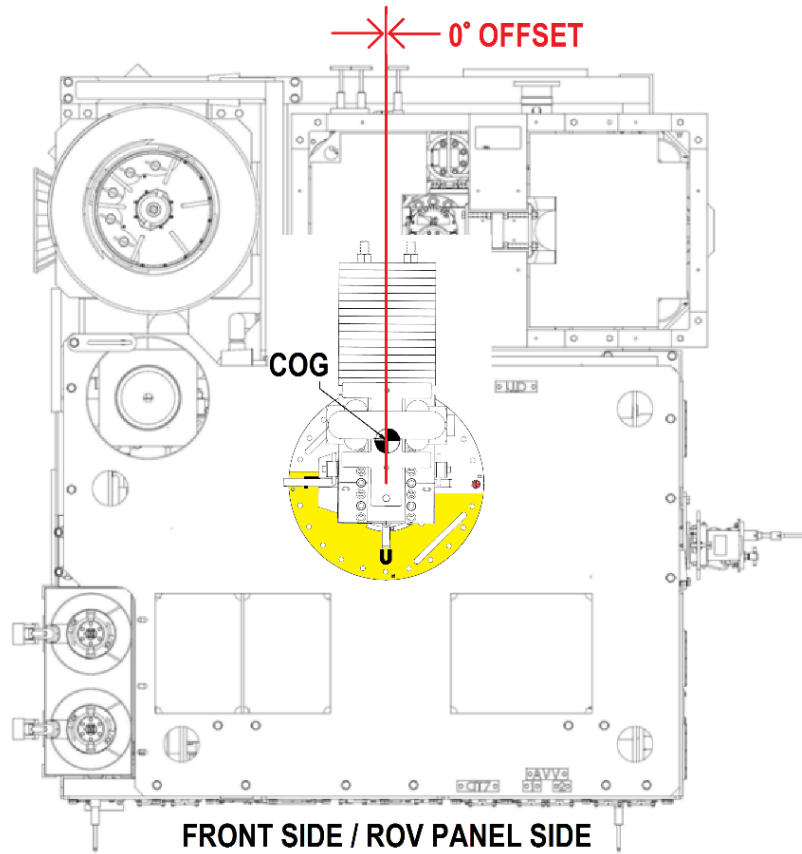


Figure 31: 0 degrees angular offset

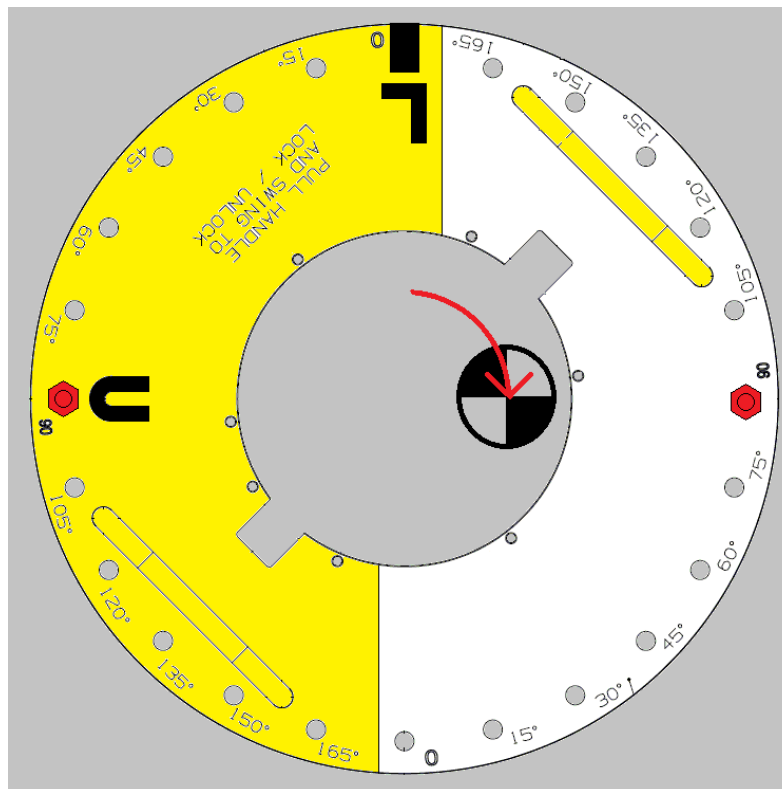
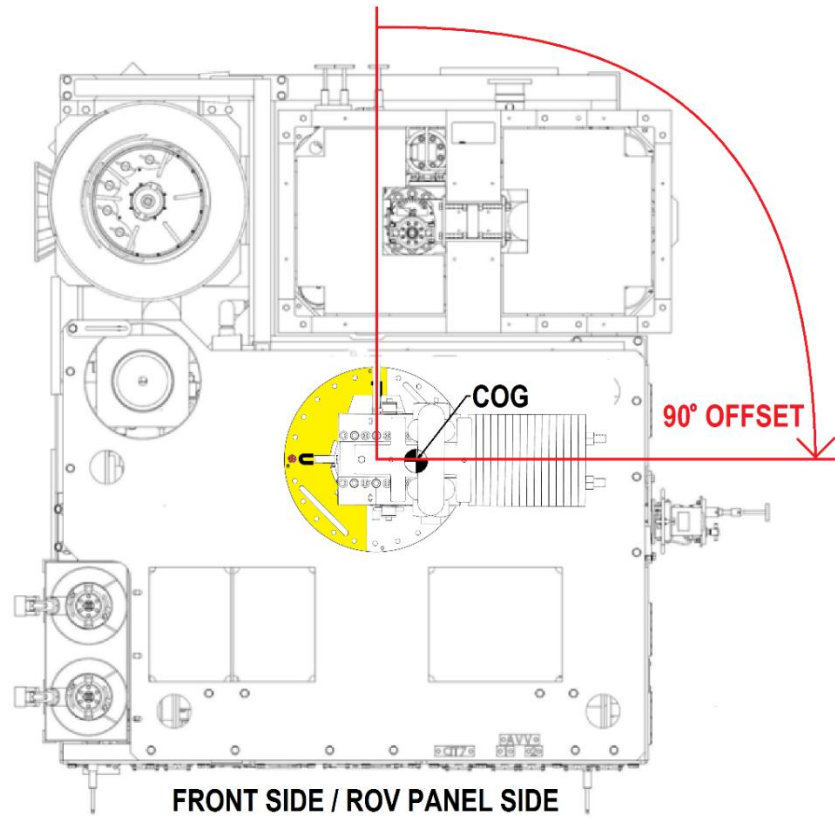
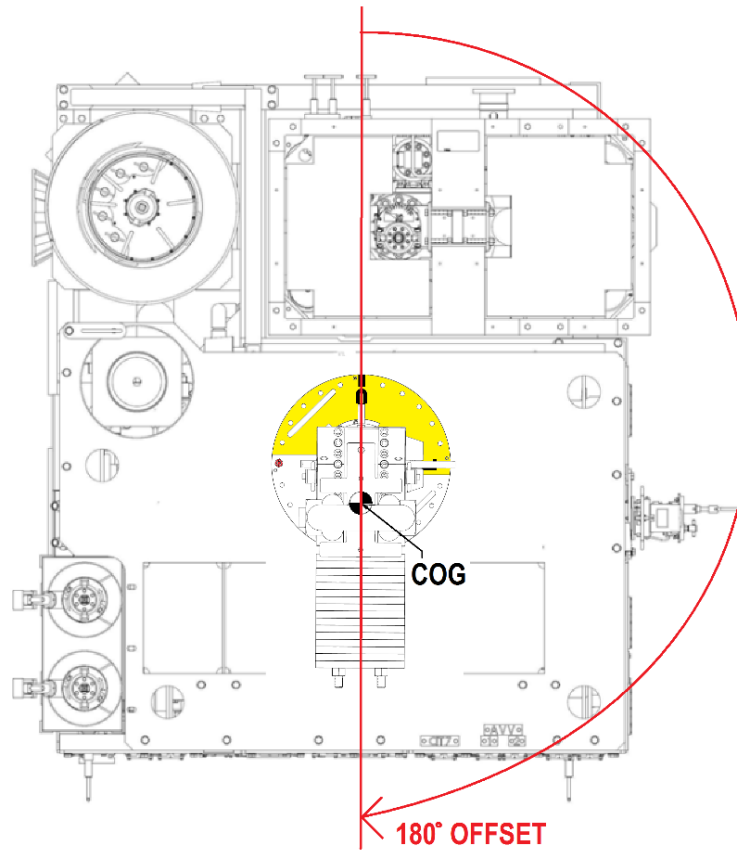


Figure 32: 90 degrees angular offset



FRONT SIDE / ROV PANEL SIDE

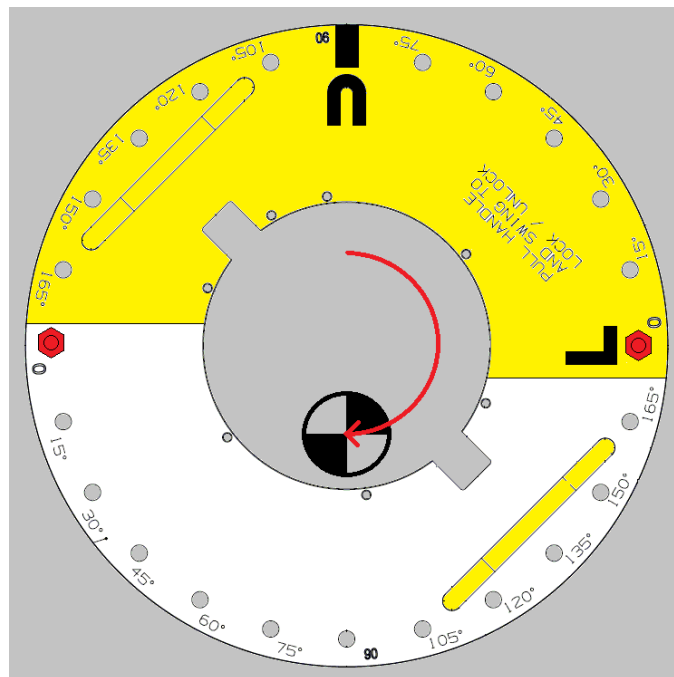


Figure 33: 180 degrees angular offset

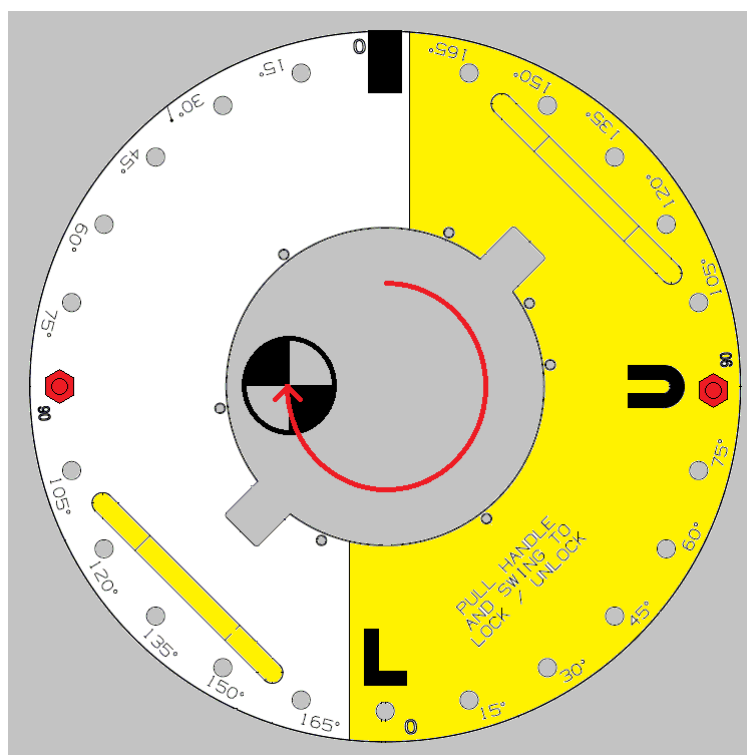
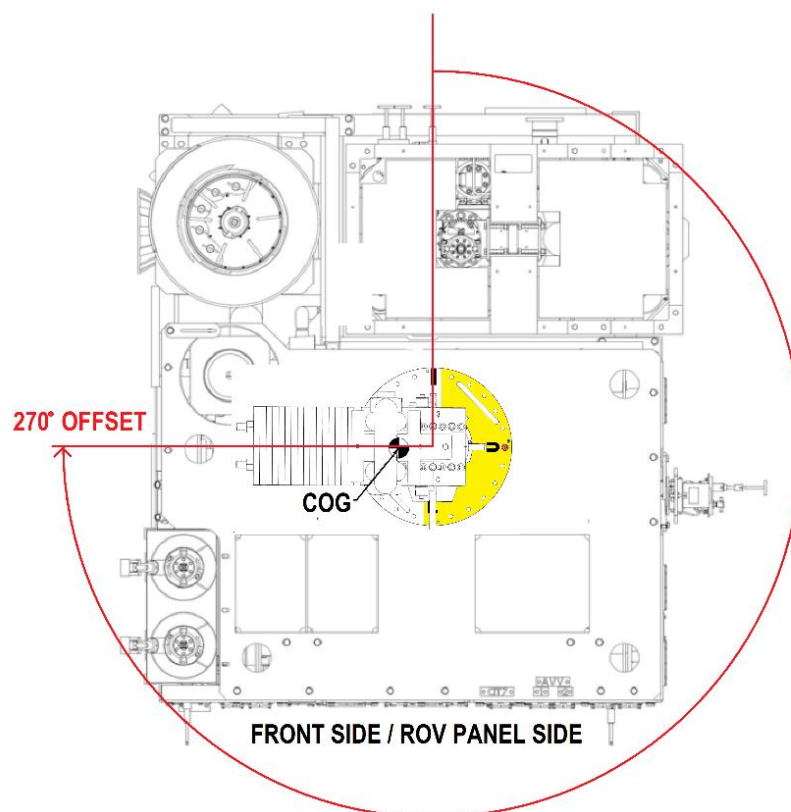


Figure 34: 270 degrees angular offset

9.2 Secondary Release Drawings

9.2.1 Secondary Release Mechanism cross section drawings

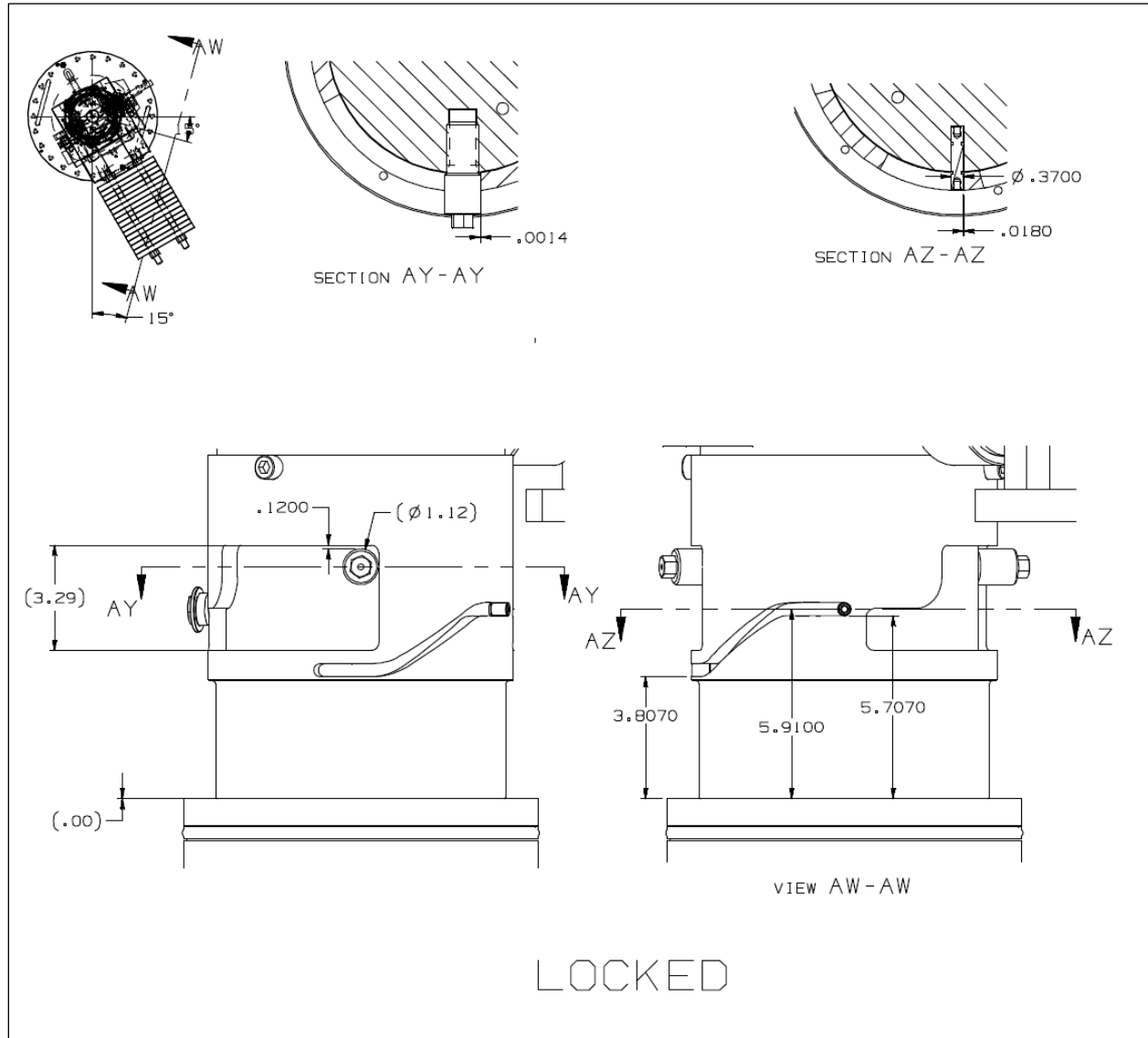


Figure 35: MTRT in Locked Position

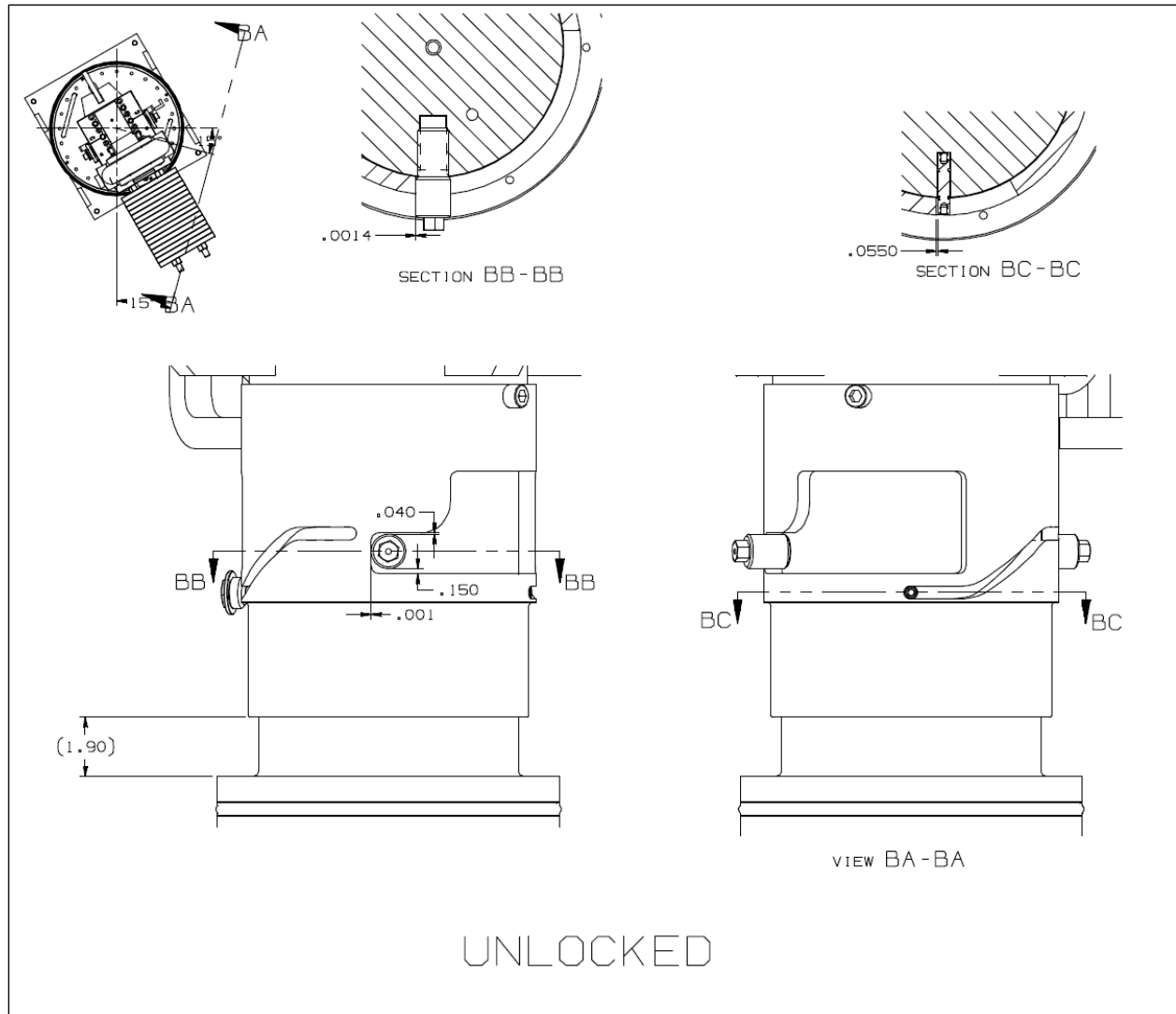


Figure 36: MTRT In Unlock Position

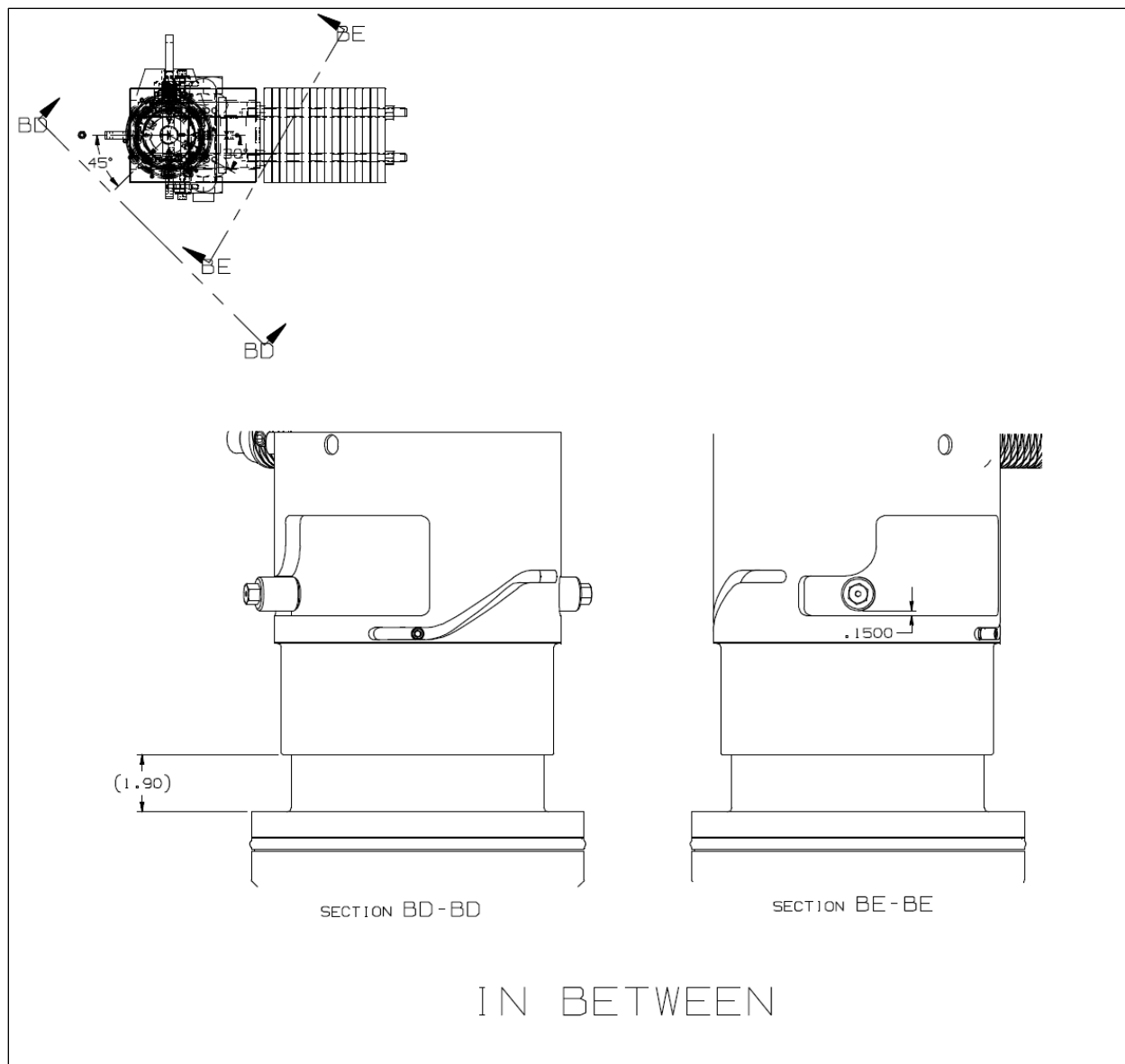


Figure 37: MTRT In Middle Position

9.2.2 Location of Secondary Release Padeyes

The figure below shows the location of the three secondary release Padeyes (green) where a tugger line may be attached to provide force to shear the Pivot Pins. A maximum of 22 kips may be applied at these locations, if required.

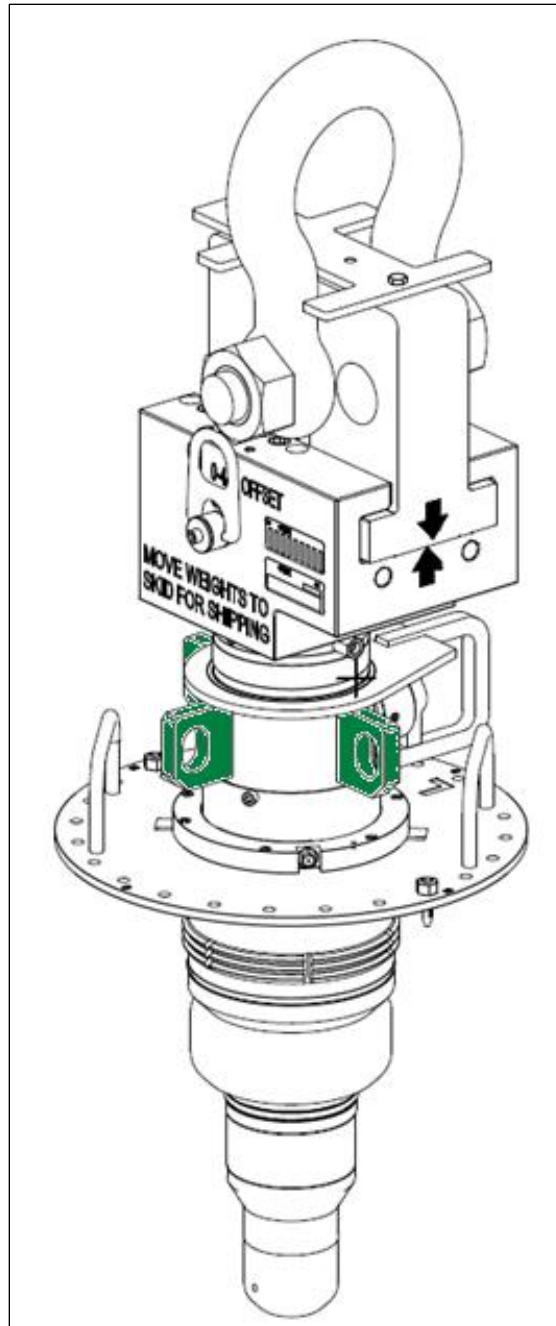


Figure 38: Secondary Release Padeyes

9.2.3 Pivot Pin Location

The figure below shows where the Pivot Pin is located (yellow arrow). Pulling the Lock Mandrel Upward will shear off Qty. 2 of these small pins and will allow the lock mandrel to move upwards.

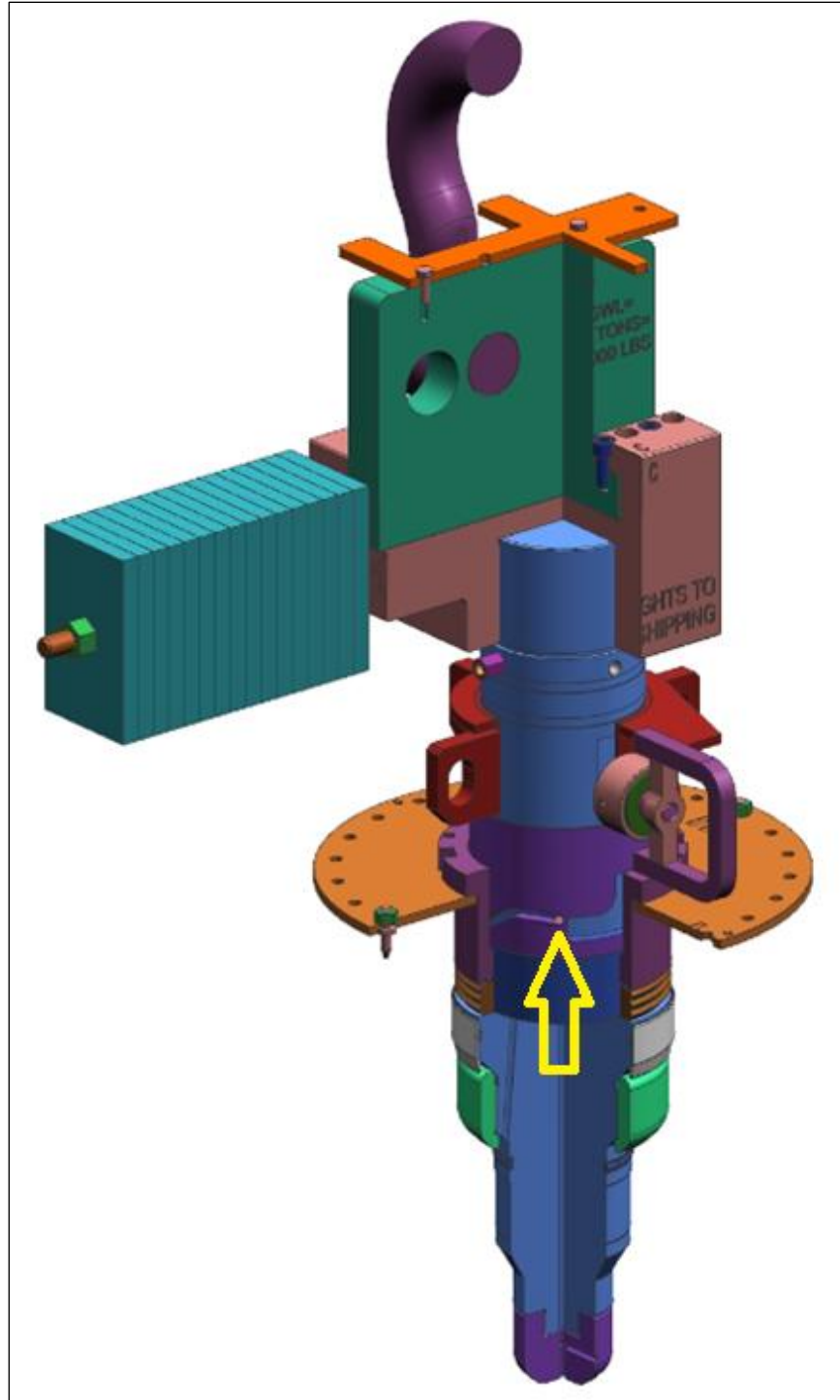


Figure 39: Pivot Pin Location

Once the Pivot Pin has sheared and the Lock Mandrel is in the unlocked position, the ROV may be used to rotate the lock mandrel. The step in the Lock Mandrel window will rest on top of the Retention Pin and hold the Lock Mandrel in the up/unlocked position. The tugger line may be removed once the tool is in the secondary unlocked and retained position. The tool can then be recovered normally using the tree sling and shackle.

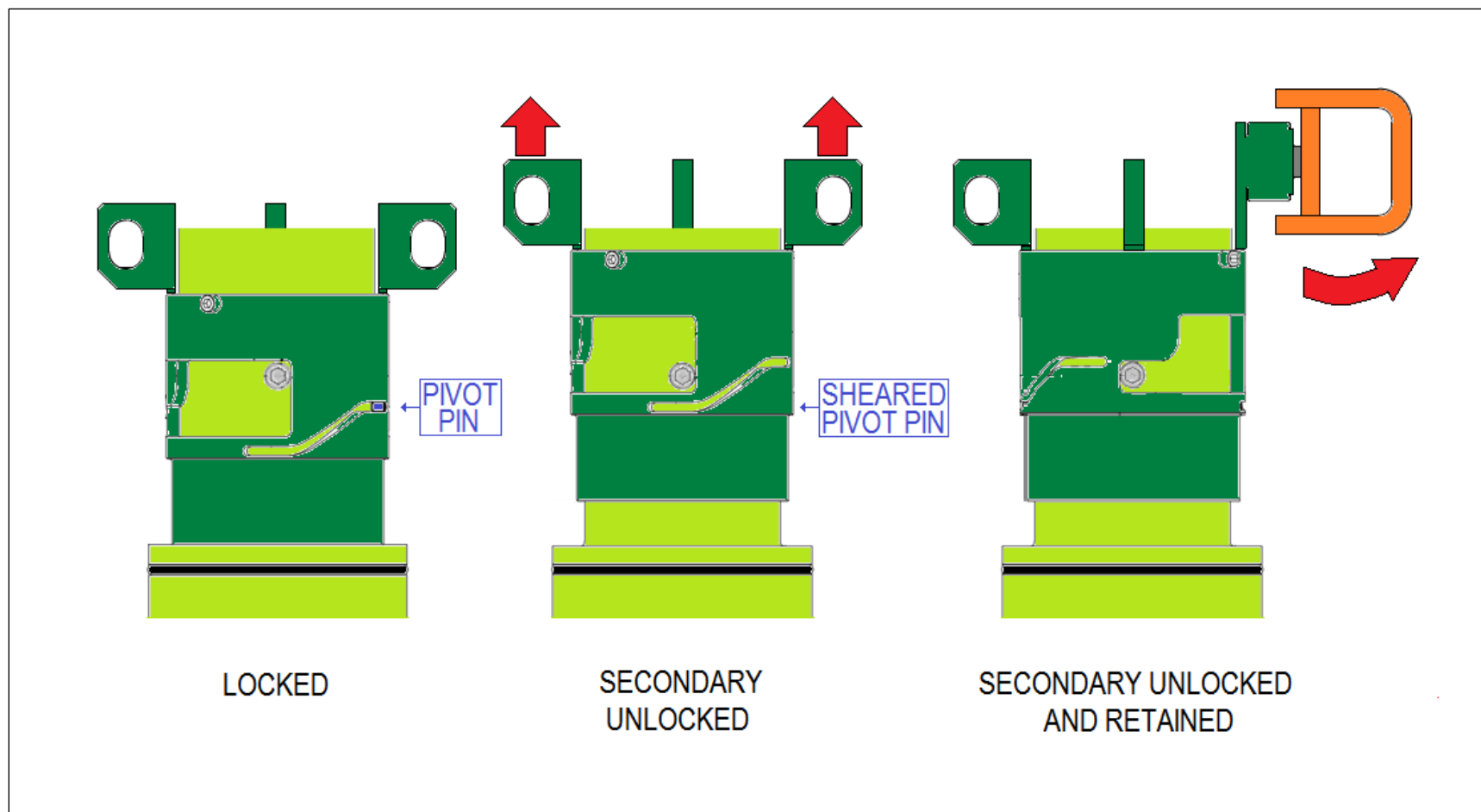


Figure 40: Secondary Unlock Sequence

